SA5#168 KSM与Cloud 逐版文字差异附录

本附录配合《本地修订核验报告》使用。每段比较标题给出前后文件名；减号为删除，加号为新增，NO TEXT CHANGE表示规范化后的可提取正文一致。所用完整原路径见《本地文件定位清单》。

比较视图包含修订插入、排除修订删除，合并空白，并在SOURCE CONTENT (TO BE DELETED)原始备份材料区之前停止。它不是图片、批注或排版差异报告。相邻版本指本地可用版本，不能由版本跳号推断其他版本不存在。无差异不意味着原Word文件字节相同。

263623逐版比较以263081为基线；另一合并来源263294的取舍在主报告中说明。

阅读方式：红色底纹和减号表示删除；绿色底纹和加号表示新增；灰色为上下文。保留全部对比内容及原文件名，不进行摘要或翻译。长行仅为适应纸张宽度自动换行。@@中的数字是规范化文本的比较行号，不是Word页码。

目录

S5-263545 - KSM1：本体、查询、共享与状态

S5-262928 KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx -> S5-263545d1 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

--- S5-262928 KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

+++ S5-263545d1 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

@@ -107,13 +107,12 @@

This key issue focuses on addressing ontology to support the knowledge management, complementing the knowledge representation defined in Key Issue #1 in clause 7.2.1, and focuses on the following aspects:

-1.Study the different types of ontology and distinguish between domain-specific ontologies (e.g., RAN ontology, CN ontology) and a cross-domain upper ontology that defines high-level concepts shared across all domains. Study how these ontologies relate to each other and to external SDOs (e.g., W3C, TM Forum, ETSI ENI, ITU-T, IEEE).

-2.How to construct the ontology for 6G management system and how to use as a shared vocabulary and a consensual formal description of knowledge. Explore how to derive ontology from existing 3GPP information models (NRM, performance measurements, alarm definitions) and from operational knowledge (e.g., expert rules, troubleshooting guides, network topology).

-3.Whether and how ontology could enable capabilities of semantics/knowledge network management, including the ability to perform deductive or inductive reasoning over ontology-conformant knowledge.

+1.Study the different types of ontology, and distinguish between a domain-independent upper ontology that defines high-level concepts shared across all domains (e.g., entity, property, constraints, relationship), domain-specific ontologies that map these concepts to specific management domains (e.g., RAN domain, CN domain), and reasoning relationships that capture semantics not present in the information models Study how these ontologies relate to each other and to external SDOs (e.g., W3C, TM Forum, ETSI ENI, ITU-T, IEEE).

+2.How to construct the ontology for 6G management system and how to use as a shared vocabulary and a consensual formal description of knowledge. Explore how to build ontology using existing 3GPP management information (NRM, performance measurements, alarm definitions) and operational knowledge (e.g., expert rules, troubleshooting guides, network topology).

+3.Whether and how ontology could enable capabilities of semantics/knowledge network management, including the ability to perform reasoning over ontology-conformant knowledge.

4.Study the possible solution sets for ontology, e.g., using RDF Schema (RDFS) and the Web Ontology Language (OWL) [x1], as defined in the W3C Semantic Web technology stack.

7.2.8.2Potential Requirements

-REQ-ONT-y1: The 6G management system shall support the definition and management of one or more ontologies that formalise knowledge.

-REQ-ONT-y2: The 6G management system shall enable ontology-based reasoning to support knowledge-based decision-making.

-REQ-ONT-y3: The ontology shall be designed to be extensible, supporting the addition of new domain-specific concepts without breaking existing dependent functions.

-REQ-ONT-y4: The 6G management system shall allow the ontology to reference and map to existing 3GPP data models (e.g., NRM, performance measurements, alarm definitions).

-REQ-ONT-y5: The 6G management system shall support ontology-based knowledge, enabling efficient query and reasoning over knowledge.

-REQ-ONT-y6: The 6G management system shall support solution sets to define ontology

+REQ-ONT-y1: The 6G management system should support the definition of a baseline ontology that formalise knowledge.

+NOTE: The baseline ontology can be extended with vendor/operator extensions to cover use-case specificities.

+REQ-ONT-y2: An otology formalizing knowledge should enable reasoning.

+REQ-ONT-y3: The 6G management system shall allow the ontology to reference and map to existing 3GPP data models (e.g., NRM, performance measurements, alarm definitions).

+REQ-ONT-y4: The 6G management system should support solution sets to define ontology

* * * Next change * * * *

@@ -123,6 +122,6 @@

8.x.a.1Potential Solution #<1>: Knowledge Scoping via context

-This solution addresses key issue 7.2.4, by defining a mechanism on knowledge MnS producer that allows a knowledge MnS consumer to specify the scope of the knowledge he wants to access (i.e., query), by providing a context to the MnS producer. This context helps the knowledge MnS producer narrow down the search space over the entire knowledge and determine which knowledge portion(s) match the scope.

+This solution addresses REQ-KNOWLEDGE-PROVISION-01 in key issue 7.2.4, by defining a mechanism on knowledge MnS producer that allows a knowledge MnS consumer to specify the knowledge he wants to access. The MnS consumer specifies this by providing a context to the MnS producer. This context helps the knowledge MnS producer narrow down the search space over the entire knowledge and determine which knowledge portion(s) match the scope.

The context that the consumer may provide include:

List of nodes the knowledge refers to. This information can be expressed with the class name (IOC) or managed object instance (MOI) representing the nodes.

-List of features the knowledge refers to. This information can be expressed with String/ENUM attribute (e.g., energy saving, capacity optimization, mobility).

+List of features the knowledge refers to. This information can be expressed with String/ENUM attribute (e.g., energy saving, capacity optimization, mobility). The criteria for feature definition is FFS.

Constraints, expressed by different dimensions, which can include:

@@ -132,5 +131,2 @@

NOTE: The list above is non-exhaustive, and can be extended with new features/key issues from further 3GPP study.

-For the knowledge MnS consumer to express the context to the knowledge MnS producer, query language can be used. The query language (e.g., SPARQL) takes care of algebraic operations when there are multiple items in the query from the producer, i.e. when the consumer specifies different information in the context.

-The knowledge portion(s) returned to the consumer is each bound to a schema identified by a unique schema namespace, which is the same namespaces that appear in the knowledge query and its response. This namespace designates the schema unambiguously and lets the consumer decide whether the returned knowledge is something it can interpret (i.e. whether it understands the vocabulary and structure).

-It is worth noting that the knowledge MnS producer decides, according to its supported capabilities, which schemas and which serialization formats it offers. Both are discoverable by the knowledge MnS consumer. If the consumer needs a schema or a format other than those offered, the normalization/translation is performed on the consumer side.

* * * Next change * * * *

@@ -139,10 +135,10 @@

This solution addresses the following key issue from Knowledge Management: key issue #6: Knowledge sharing (clause 7.2.6).

-This solution defines the concept of knowledge sharing group as the mechanism through which multiple knowledge MnS producers collaboratively contribute to shared knowledge instances. A sharing group has the following characteristics:

-A sharing group has one or more members. Each member is a knowledge MnS producer.

+This solution defines the concept of knowledge sharing group as the mechanism through which multiple entities collaboratively contribute to shared knowledge instances. A sharing group has the following characteristics:

+A sharing group consists of the following members: one knowledge MnS producer and multiple knowledge MnS consumers.

A sharing group is associated with one or more shared knowledge instances. Each shared knowledge is associated with exactly one sharing group.

Membership in a sharing group grants the member the ability to contribute to the shared knowledge instances associated with the group.

-Members of the sharing group can observe each other’ s accepted contributions (i.e. contributions that have been validated and merged) only when the contributor has posted the knowledge to the knowledge repository.

+Members of the sharing group can observe each other’ s contributions.

A knowledge sharing group supports the following lifecycle operations:

Creation of the sharing group with an initial member list and associated knowledge instances.

-Addition (removal) of a knowledge MnS producer to (from) the member list.

+Addition (removal) of members to (from) the group.

Association (disassociation) of a new shared knowledge instance with (from) the group.

@@ -151,12 +147,10 @@

sharingGroupId: it is unique identifier of the sharing group.

-memberList: distinguished names of knowledge MnS producers that are members of the group.

+memberList: distinguished names of entities that are members of the group.

knowledgeInstanceRefList: references to the shared knowledge instances associated with this group.

-A shared knowledge instance belongs to one single shared group. The contributions submitted into a shared knowledge instance need to be access controlled (ensures the contribution comes from a MnS producer which is part of the group) and validated (ensures the contribution does not compromise integrity of the knowledge) before getting accepted. These two activities are inherent to the knowledge authority role. A shared knowledge instance can have one or more knowledge authorities (see NOTE 1, NOTE 2).

-NOTE 1: A knowledge consisting of “N” non-overlapping portions can have at most “N” knowledge authorities. This means that very non-overlapping portion cannot have more than one designated authority, i.e. there is exactly one authority responsible for accounting validation.

-NOTE 2: A knowledge authority can be performed by one of the MnS producers within the group, or by an external entity. How this selection is done is FFS at this stage.

-The shared knowledge instance will include the same attributes of a knowledge instance plus two other:

-sharingGroupRef: identifier of the sharing group where the knowledge instance belongs to.

-authorityRefList: distinguished name of knowledge MnS producer(s) taking the authority role.

-The workflow below illustrates an example of a sharing group consisting of three knowledge MnS producers: A, B, C. All of them are contributors. The group only has one single shared knowledge instance, which is knowledge_1. The MnS producer C holds the authority role for this shared knowledge instance. With this setup, the workflow solution looks like as below:

+A shared knowledge instance belongs to one single shared group. The contributions submitted into a shared knowledge instance need to be access controlled (ensures the contribution comes from a MnS producer which is part of the group) and validated (ensures the contribution does not compromise integrity of the knowledge) before getting accepted.

+The workflow below illustrates an example of a sharing group consisting of three members: one knowledge MnS producer, and two knowledge MnS consumersThe group only has one single shared knowledge, which is knowledge_1. Each knowledge MnS consumer within the group can write contributions to knowledge 1, and read contributions made to knowledge_1.With this setup, the workflow solution looks like as below:

Figure 8.x.y: Knowledge sharing workflow sequence

+When knowledge MnS consumer A contributes to knowledge_1, it sends contribution to knowledge MnS producer (step 1). If the consumer A is part of the sharing group and the contribution is valid, then the producer stores it in a knowledge base. Following this, the producer sends a response back to the knowledge MnS consumer A informing about this (step 2), and also notifies the rest of knowledge MnS consumers members of the sharing group that a contribution has been made to knowledge_1 (step 3).

+When knowledge MnS consumer B makes a contribution to knowledge_1, same procedure is applied (steps 4-6).

+When a knowledge MnS consumer C makes a contribution to knowledge_1 (step 7), the knowledge MnS producer identifies that is not a part of the sharing group, so its contribution is not accepted. The producer only sends a response back to the knowledge MnS consumer C informing about this (step 8).

* * * Next change * * * *

@@ -166,46 +160,9 @@

The present solution addresses the first aspect, by providing a knowledge state model. This model defines the following states.

-CREATED: The knowledge instance has been generated/created, but has not been validated yet. Therefore, the knowledge instance is not yet accessible for consumption by knowledge MnS consumers.

ACTIVE: The knowledge instance is available for consumption by authorized knowledge MnS consumers. This is the normal operational state in which knowledge can be discovered, queried, and retrieved.

SUSPENDED: The knowledge instance is currently unavailable for consumption due to one or more transient conditions. Once the conditions are resolved, the knowledge instance can be set to ACTIVE state.

-DELETED: The knowledge instance is removed from the active knowledge base and is no longer accessible for consumption. The knowledge may optionally be archived for audit purposes but is not discoverable or retrievable.

+DELETED: The knowledge instance is removed and is no longer accessible for consumption.

Figure 8.x.y: State model for knowledge instances.

-Every knowledge instance has a property/attribute specifying whether it is valid or not. For a knowledge instance to be made accessible for consumption, it needs to be valid. The validity of a knowledge instance in relation to the lifecycle state is as follows. When in CREATED or DELETED, the knowledge instance is not valid. When in ACTIVE, the knowledge instance is valid. When in SUSPENDED, the knowledge instance can be either valid (this means the transient condition is an accessibility issue, e.g. the knowledge MnS producer is currently unavailable) or not valid (this means the transient condition is a validity issue, e.g. detection of knowledge conflict, knowledge re-validation needed, etc.).

-Table 8.x.y illustrates the knowledge states along with allowed transitions. The table below provides example triggering conditions for these transitions.

-Transition

-Source state

-Target State

-Triggering conditions

-1

-CREATED

-ACTIVE

-Successful knowledge validation

-Explicit change by knowledge authority.

-2

-ACTIVE

-SUSPENDED

-Knowledge MnS producer / authority unavailability.
-Network change (e.g., a change in the network that affects the validity of the knowledge).
-Detection of knowledge conflict (conflict resolution needed).
-Re-validation needed.
-Explicit change by knowledge authority.
-3
-SUSPENDED
-ACTIVE
-Knowledge MnS producer /authority recovery.
-Successful knowledge re-validation.
-Explicit change by knowledge authority.
-4
-SUSPENDED
-DELETED
-Explicit change by knowledge authority
-5
-ACTIVE
-DELETED
-Explicit change by knowledge authority.
-6
-CREATED
-DELETED
-Explicit change by knowledge authority
-Failed knowledge validation (after certain number of attempts).
-Table 8.x.y: Examples of triggering conditions for state transitions.
+Every knowledge instance has a property/attribute specifying whether it is valid or not. The validity of a knowledge instance in relation to the lifecycle state is as follows. When in ACTIVE, the knowledge instance is valid. When in SUSPENDED, the knowledge instance can be either valid (this means the transient condition is an accessibility issue, e.g. the knowledge MnS producer is currently unavailable) or not valid (this means the transient condition is a validity issue, e.g. detection of knowledge conflict, knowledge re-validation needed, etc.). When in DELETED, the knowledge instance is not valid.
+It can happen that a knowledge MnS consumer accessed a knowledge instance when in ACTIVE state, but later knowledge instance transitions to SUSPENDED or DELETED. In that case, the knowledge MnS consumer needs to be flagged about this, to enable an informed consumption.
+The transitions across the identified states is FFS.
* * * Next change * * * *
@@ -221,3 +178,3 @@

The above approach allows modular ontology construction: a governed baseline ontology (levels 1-3) that can be extended with proprietary artifacts (Level 4). Level 1-3 forms the standardized ontology. Application ontologies can be defined by composing elements from Levels 1–3 into use-case-specific schemas that define which entities, relationships, and constraints are relevant for a particular management domain or feature (e.g., energy saving, mobility optimization, spectrum management). Application ontologies enable interoperable knowledge exchange within a particular management scenario or application.

-Ontologies in 6G management ca be maintained and versioned. This means that an ontology may be updated to accommodate network evolution and new management concepts, with version control ensuring interoperability across management entities.

+Ontologies in 6G management can be maintained and versioned. This means that an ontology may be updated to accommodate network evolution and new management concepts, with version control ensuring interoperability across management entities.

The representation of the knowledge, including its mapping to existing semantic technology standards, is recognized and recommended as the reference baseline but is not re-elaborated beyond the recommendation. The recommended representation baseline reuses RDF/RDFS as data model, OWL for formal ontology semantics, SHACL for constraints and SPARQL as reference query surface, with IRI/URI identification and namespaces. Reuse of existing external ontologies (e.g., SAREF, NGSI-LD / smart-data-models, TM Forum ontologies) is recommended for domain modules, aligned via standard OWL/RDFS constructs.

S5-263545d1 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx -> S5-263545d2 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

--- S5-263545d1 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

+++ S5-263545d2 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

@@ -121,12 +121,10 @@

8.x.a Potential Solutions for Key Issue #4: Knowledge Provision

-8.x.a.1Potential Solution #<1>: Knowledge Scoping via context

-This solution addresses REQ-KNOWLEDGE-PROVISION-01 in key issue 7.2.4, by defining a mechanism on knowledge MnS producer that allows a knowledge MnS consumer to specify the knowledge he wants to access. The MnS consumer specifies this by providing a context to the MnS producer. This context helps the knowledge MnS producer narrow down the search space over the entire knowledge and determine which knowledge portion(s) match the scope.

-The context that the consumer may provide include:

-List of nodes the knowledge refers to. This information can be expressed with the class name (IOC) or managed object instance (MOI) representing the nodes.

-List of features the knowledge refers to. This information can be expressed with String/ENUM attribute (e.g., energy saving, capacity optimization, mobility). The criteria for feature definition is FFS.

-Constraints, expressed by different dimensions, which can include:

-Area-based constraints. Baseline solution could be the use of GeoArea <<datatype>> (TS 28.622).

-Time-based constraints. Baseline solution could be the use of TimeWindow <<datatype>> (TS 28.622).

-Quality-based constraints. Examples of these constraints could be knowledge confidence, expressed as Integer (e.g., 0-100), Float (e.g., 0-1) or String/ENUM (e.g., high, medium, low).

-NOTE: The list above is non-exhaustive, and can be extended with new features/key issues from further 3GPP study.

+8.x.a.1Potential Solution #<1>: Knowledge Query via context

+This solution addresses REQ-KNOWLEDGE-PROVISION-01 in key issue 7.2.4, by defining a mechanism on knowledge MnS producer that allows a knowledge MnS consumer to specify the knowledge he wants to access. The MnS consumer specifies this by providing a context to the MnS producer. This context helps the knowledge MnS producer determine which knowledge portion(s) match the context.

+The context that the consumer may provide include information related to:

+Knowledge scope, specifying what the knowledge is applicable to. Examples include geographic area, list of network entities, network domain, features (e.g., energy saving, mobility, etc).

+Knowledge quality (e.g., confidence).

+Knowledge generation (e.g., generation time).

+NOTE 1: The list above is non-exhaustive, and can be extended with new features/key issues from further 3GPP study.

+NOTE 2: Baseline solutions for the listed examples is FFS.

* * * Next change * * * *

@@ -164,3 +162,3 @@

Figure 8.x.y: State model for knowledge instances.

-Every knowledge instance has a property/attribute specifying whether it is valid or not. The validity of a knowledge instance in relation to the lifecycle state is as follows. When in ACTIVE, the knowledge instance is valid. When in SUSPENDED, the knowledge instance can be either valid (this means the transient condition is an accessibility issue, e.g. the knowledge MnS producer is currently unavailable) or not valid (this means the transient condition is a validity issue, e.g. detection of knowledge conflict, knowledge re-validation needed, etc.). When in DELETED, the knowledge instance is not valid.

+Every knowledge instance has a property/attribute specifying whether it is valid or not (i.e. marked as suspected invalid). The validity of a knowledge instance in relation to the lifecycle state is as follows. When in ACTIVE, the knowledge instance is valid. When in SUSPENDED, the knowledge instance can be either valid (this means the transient condition is an accessibility issue, e.g. the knowledge MnS producer is currently unavailable) or not valid (this means the transient condition is a validity issue, e.g. detection of knowledge conflict, knowledge re-validation needed, etc.). When in DELETED, the knowledge instance is not valid.

It can happen that a knowledge MnS consumer accessed a knowledge instance when in ACTIVE state, but later knowledge instance transitions to SUSPENDED or DELETED. In that case, the knowledge MnS consumer needs to be flagged about this, to enable an informed consumption.

@@ -172,11 +170,10 @@

This solution aims to define a 3GPP-based ontology for knowledge. This ontology serves as the semantic foundation for knowledge representation and management in the 6G management system. This ontology is structured in four levels of increasing specificity:

-Level 1 (Top-level ontology). This level comprises the definition of a domain-independent core ontology. It specifies a small, stable set of abstract concepts (such as Entity, Relationship, Property, Event, Constraint, Action, Effect, Context, Time) and their structural relationships, providing a shared vocabulary that is independent of any specific network domain. The governance of this level is for further study; options include defining it normatively within 3GPP, adopting or profiling an existing ontology from another SDO, or co-developing it with a partner SDO.

-Level 2 (domain-level ontology). This level comprises the definition of a domain-specific core ontology. It specifies a formal mapping that instantiates the domain-independent core concepts against existing 3GPP management information, including NRM (IOCs, attributes and structural relationships like name-containment, association) and management data (e.g., performance measurements, KPIs, analytics, alarm definitions, etc.). This level frames this 3GPP management information into an ontology format.

-Level 3 (Reasoning-level ontology). This level specifies additional semantically typed relationships not present in the NRM but required to enable richer reasoning, for example: causality (e.g., causedBy/affects), conflict detection (e.g., conflictsWith), provenance (e.g., derivedFrom), temporal validity (e.g., validDuring), versioning (e.g., supersedes). These relationships enable impact analysis, root-cause reasoning and conflict detection across domains.

+Level 1 (Top-level ontology). This level comprises the definition of a domain-independent ontology. It specifies a small, stable set of abstract concepts (such as Entity, Relationship, Property, Event, Constraint, Action, Effect, Context, Time) and their structural relationships, providing a shared vocabulary that is independent of any specific network domain. The governance of this level is for further study; options include defining it normatively within 3GPP, adopting or profiling an existing ontology from another SDO, or co-developing it with a partner SDO.

+Level 2 (domain-level ontology). This level comprises the definition of a domain-specific ontology. It specifies a formal mapping that instantiates the domain-independent core concepts against existing 3GPP management information, including NRM (IOCs, attributes and structural relationships like name-containment, association) and management data (e.g., performance measurements, KPIs, analytics, alarm definitions, etc.). This level frames this 3GPP management information into an ontology format.

+Level 3 (Reasoning-level ontology). This level specifies additional semantically typed relationships not present in the NRM but required to enable richer reasoning, for example: causality (e.g., causedBy/affects), conflict detection (e.g., conflictsWith), versioning (e.g., supersedes). These relationships enable impact analysis, root-cause reasoning and conflict detection across domains.

Level 4 (Extension-level ontology). This level comprises a governed extension mechanism allowing vendors and operators to introduce proprietary concepts and relationships (e.g., vendor-specific causal models between configuration parameters and KPIs, proprietary hardware component relationships) on top of previous levels, without breaking interoperability.

-Meta-knowledge (provenance, confidence, temporal validity) is treated as first-class at all levels, enabling knowledge MnS consumers to assess trustworthiness before acting.

-The above approach allows modular ontology construction: a governed baseline ontology (levels 1-3) that can be extended with proprietary artifacts (Level 4). Level 1-3 forms the standardized ontology. Application ontologies can be defined by composing elements from Levels 1–3 into use-case-specific schemas that define which entities, relationships, and constraints are relevant for a particular management domain or feature (e.g., energy saving, mobility optimization, spectrum management). Application ontologies enable interoperable knowledge exchange within a particular management scenario or application.

-Ontologies in 6G management can be maintained and versioned. This means that an ontology may be updated to accommodate network evolution and new management concepts, with version control ensuring interoperability across management entities.

-The representation of the knowledge, including its mapping to existing semantic technology standards, is recognized and recommended as the reference baseline but is not re-elaborated beyond the recommendation. The recommended representation baseline reuses RDF/RDFS as data model, OWL for formal ontology semantics, SHACL for constraints and SPARQL as reference query surface, with IRI/URI identification and namespaces. Reuse of existing external ontologies (e.g., SAREF, NGSI-LD / smart-data-models, TM Forum ontologies) is recommended for domain modules, aligned via standard OWL/RDFS constructs.

-Reasoning engines, inference algorithms, concrete wire serializations and internal tooling are left to implementation. The standard defines the shared vocabulary, its grounding, its meta-knowledge model and its interface exposure; it does not prescribe how inference is executed or how the model is internally encoded.

+The above approach allows modular ontology construction: a governed baseline ontology (levels 1-3) that can be extended with proprietary artifacts (Level 4). Level 1-3 forms the standardized base ontology. Application ontologies can be defined by composing elements from Levels 1–3 into use-case-specific schemas that define which entities, relationships, and constraints are relevant for a particular management domain or feature (e.g., energy saving, mobility optimization, spectrum management). Application ontologies enable interoperable knowledge exchange within a particular management scenario or application.

+Ontologies in 6G management can be maintained and versioned. This means that an ontology may be updated to accommodate network evolution and new management concepts, with version control ensuring interoperability.

+The representation of the knowledge, including its mapping to existing semantic technology standards, is recognized and recommended as the reference baseline but is not re-elaborated beyond the recommendation. The recommended representation baseline reuses RDF/RDFS as data model, OWL for formal ontology semantics, SHACL for constraints and

SPARQL as reference query surface, with IRI/URI identification and namespaces. Existing external ontologies (e.g., SAREF, NGSI-LD / smart-data-models, TM Forum ontologies) can be reused when needed.

+Reasoning engines, inference algorithms, concrete wire serializations and internal tooling are left to implementation. The standard defines the shared vocabulary¡ and its interface exposure; it does not prescribe how inference is executed or how the model is internally encoded.

* * * End of Changes * * * *

S5-263545d2 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx -> S5-263545d4 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

--- S5-263545d2 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

+++ S5-263545d4 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

@@ -161,3 +161,2 @@

DELETED: The knowledge instance is removed and is no longer accessible for consumption.

-Figure 8.x.y: State model for knowledge instances.

Every knowledge instance has a property/attribute specifying whether it is valid or not (i.e. marked as suspected invalid). The validity of a knowledge instance in relation to the lifecycle state is as follows. When in ACTIVE, the knowledge instance is valid. When in SUSPENDED, the knowledge instance can be either valid (this means the transient condition is an accessibility issue, e.g. the knowledge MnS producer is currently unavailable) or not valid (this means the transient condition is a validity issue, e.g. detection of knowledge conflict, knowledge re-validation needed, etc.). When in DELETED, the knowledge instance is not valid.

@@ -174,2 +173,3 @@

Level 4 (Extension-level ontology). This level comprises a governed extension mechanism allowing vendors and operators to introduce proprietary concepts and relationships (e.g., vendor-specific causal models between configuration parameters and KPIs, proprietary hardware component relationships) on top of previous levels, without breaking interoperability.

+NOTE: The numerology adopted for levels is simply a convention.

The above approach allows modular ontology construction: a governed baseline ontology (levels 1-3) that can be extended with proprietary artifacts (Level 4). Level 1-3 forms the standardized base ontology. Application ontologies can be defined by composing elements from Levels 1–3 into use-case-specific schemas that define which entities, relationships, and constraints are relevant for a particular management domain or feature (e.g., energy saving, mobility optimization, spectrum management). Application ontologies enable interoperable knowledge exchange within a particular management scenario or application.

S5-263545d4 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx -> S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

--- S5-263545d4 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

+++ S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

@@ -1,3 +1,3 @@

-3GPP TSG-SA5 Meeting #168S5-262928

-Prague, Czech Republic, 24 – 28 Aug 2026

+3GPP TSG-SA5 Meeting #168S5-263545

+Prague, Czech Republic, 24 – 28 Aug 2026Revision of S5-262928

Source:Ericsson (moderator)

@@ -130,26 +130,2 @@

* * * Next change * * * *

-8.x.b Potential Solutions for Key Issue #6: Knowledge Sharing

-8.x.b.1Potential Solution #<1>: Knowledge sharing

-This solution addresses the following key issue from Knowledge Management: key issue #6: Knowledge sharing (clause 7.2.6).

-This solution defines the concept of knowledge sharing group as the mechanism through which multiple entities collaboratively contribute to shared knowledge instances. A sharing group has the following characteristics:

-A sharing group consists of the following members: one knowledge MnS producer and multiple knowledge MnS consumers.

-A sharing group is associated with one or more shared knowledge instances. Each shared knowledge is associated with exactly one sharing group.

-Membership in a sharing group grants the member the ability to contribute to the shared knowledge instances associated with the group.

-Members of the sharing group can observe each other' s contributions.

-A knowledge sharing group supports the following lifecycle operations:

-Creation of the sharing group with an initial member list and associated knowledge instances.

-Addition (removal) of members to (from) the group.

-Association (disassociation) of a new shared knowledge instance with (from) the group.

-Deletion of the sharing group.

-A sharing group can be modelled with an information element comprising the following attributes:

-sharingGroupId: it is unique identifier of the sharing group.

-memberList: distinguished names of entities that are members of the group.

-knowledgeInstanceRefList: references to the shared knowledge instances associated with this group.

-A shared knowledge instance belongs to one single shared group. The contributions submitted into a shared knowledge instance need to be access controlled (ensures the contribution comes from a MnS producer which is part of the group) and validated (ensures the contribution does not compromise integrity of the knowledge) before getting accepted.

-The workflow below illustrates an example of a sharing group consisting of three members: one knowledge MnS producer, and two knowledge MnS consumersThe group only has one single shared knowledge, which is knowledge_1. Each knowledge MnS consumer within the group can write contributions to knowledge 1, and read contributions made to knowledge_1.With this setup, the workflow solution looks like as below:

-Figure 8.x.y: Knowledge sharing workflow sequence

-When knowledge MnS consumer A contributes to knowledge_1, it sends contribution to knowledge MnS producer (step 1). If the consumer A is part of the sharing group and the contribution is valid, then the producer stores it in a knowledge base. Following this, the producer sends a response back to the knowledge MnS consumer A informing about this (step 2), and also notifies the rest of knowledge MnS consumers members of the sharing group that a contribution has been made to knowledge_1 (step 3).

-When knowledge MnS consumer B makes a contribution to knowledge_1, same procedure is applied (steps 4-6).

-When a knowledge MnS consumer C makes a contribution to knowledge_1 (step 7), the knowledge MnS producer identifies that is not a part of the sharing group, so its contribution is not accepted. The producer only sends a response back to the knowledge MnS consumer C informing about this (step 8).

-* * * Next change * * * *

8.x.c Potential Solutions for Key Issue #7: Knowledge Lifecycle Management

@@ -166,14 +142,9 @@

8.x.d Potential Solutions for Key Issue #8: Ontology

-8.2.d.1 Potential Solution #<1>: Knowledge Ontology and Representation

-This solution addresses key issue #8: "Ontology" (clause 7.2.8) and relates to key issue #1: "Knowledge Representation and Management" (clause 7.2.1).

-This solution aims to define a 3GPP-based ontology for knowledge. This ontology serves as the semantic foundation for knowledge representation and management in the 6G management system. This ontology is structured in four levels of increasing specificity:

-Level 1 (Top-level ontology). This level comprises the definition of a domain-independent ontology. It specifies a small, stable set of abstract concepts (such as Entity, Relationship, Property, Event, Constraint, Action, Effect, Context, Time) and their structural relationships, providing a shared vocabulary that is independent of any specific network domain. The governance of this level is for further study; options include defining it normatively within 3GPP, adopting or profiling an existing ontology from another SDO, or co-developing it with a partner SDO.

-Level 2 (domain-level ontology). This level comprises the definition of a domain-specific ontology. It specifies a formal mapping that instantiates the domain-independent core concepts against existing 3GPP management information, including NRM (IOCs, attributes and structural relationships like name-containment, association) and management data (e.g., performance measurements, KPIs, analytics, alarm definitions, etc.). This level frames this 3GPP management information into an ontology format.

-Level 3 (Reasoning-level ontology). This level specifies additional semantically typed relationships not present in the NRM but required to enable richer reasoning, for example: causality (e.g., causedBy/affects), conflict detection (e.g., conflictsWith), versioning (e.g., supersedes). These relationships enable impact analysis, root-cause reasoning and conflict detection across domains.

-Level 4 (Extension-level ontology). This level comprises a governed extension mechanism allowing vendors and operators to introduce proprietary concepts and relationships (e.g., vendor-specific causal models between configuration parameters and KPIs, proprietary hardware component relationships) on top of previous levels, without breaking interoperability.

-NOTE: The numerology adopted for levels is simply a convention.

-The above approach allows modular ontology construction: a governed baseline ontology (levels 1-3) that can be extended with proprietary artifacts (Level 4). Level 1-3 forms the standardized base ontology. Application ontologies can be defined by composing elements from Levels 1–3 into use-case-specific schemas that define which entities, relationships, and constraints are relevant for a particular management domain or feature (e.g., energy saving, mobility optimization, spectrum management). Application ontologies enable interoperable knowledge exchange within a particular management scenario or application.

-Ontologies in 6G management can be maintained and versioned. This means that an ontology may be updated to accommodate network evolution and new management concepts, with version control ensuring interoperability.

-The representation of the knowledge, including its mapping to existing semantic technology standards, is recognized and recommended as the reference baseline but is not re-elaborated beyond the recommendation. The recommended representation baseline reuses RDF/RDFS as data model, OWL for formal ontology semantics, SHACL for constraints and SPARQL as reference query surface, with IRI/URI identification and namespaces. Existing external ontologies (e.g., SAREF, NGSI-LD / smart-data-models, TM Forum ontologies) can be reused when needed.

-Reasoning engines, inference algorithms, concrete wire serializations and internal tooling are left to implementation. The standard defines the shared vocabulary¡ and its interface exposure; it does not prescribe how inference is executed or how the model is internally encoded.

+8.2.d.1 Potential Solution #<1>: Knowledge Ontology

+This solution addresses key issue #8: "Ontology" (clause 7.2.8).

+This solution aims to define a 3GPP-based ontology for knowledge. This ontology serves as the semantic foundation for knowledge representation and management in the 6G management system. This ontology shall include the following aspects:

+Formalize 3GPP management information into ontology format. This includes the definition of a domain-independent ontology that specifies a small, stable set of abstract concepts (such as Entity, Relationship, Property, Event, Constraint, Action, Effect, Context, Time) and their structural relationships, providing a shared vocabulary that is independent of any specific network domain. This also comprises the specification of formal mapping that instantiates the domain-independent core concepts against existing 3GPP management information, including NRM (IOCs, attributes and structural relationships like name-containment, association) and management data (e.g., performance measurements, KPIs, analytics, alarm definitions, etc.).

+Specifying additional semantically typed relationships that are not present in the NRM but required to enable richer reasoning, for example: causality (e.g., causedBy/affects), conflict detection (e.g., conflictsWith), versioning (e.g., supersedes). These relationships enable impact analysis, root-cause reasoning and conflict detection across domains.

+The above approach allows modular ontology construction. Ontologies in 6G management can be maintained and versioned. This means that an ontology may be updated to accommodate network evolution and new management concepts, with version control ensuring interoperability.

+Reasoning engines, inference algorithms, concrete wire serializations and internal tooling are left to implementation. The standard defines the shared vocabulary and its interface exposure; it does not prescribe how inference is executed or how the model is internally encoded.

* * * End of Changes * * * *

S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx -> S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online.docx

--- S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

+++ S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online.docx

@@ -114,5 +114,4 @@

NOTE: The baseline ontology can be extended with vendor/operator extensions to cover use-case specificities.

-REQ-ONT-y2: An otology formalizing knowledge should enable reasoning.

-REQ-ONT-y3: The 6G management system shall allow the ontology to reference and map to existing 3GPP data models (e.g., NRM, performance measurements, alarm definitions).

-REQ-ONT-y4: The 6G management system should support solution sets to define ontology

+REQ-ONT-y2: The 6G management system should support definition of ontology reasoning rules for knowledge.

+REQ-ONT-y3: The 6G management system should support the ontology to reference and map to existing 3GPP data models (e.g., NRM, performance measurements, alarm definitions).

* * * Next change * * * *

S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online.docx -> S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online1.docx

--- S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online.docx

+++ S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online1.docx

@@ -114,4 +114,3 @@

NOTE: The baseline ontology can be extended with vendor/operator extensions to cover use-case specificities.

-REQ-ONT-y2: The 6G management system should support definition of ontology reasoning rules for knowledge.

-REQ-ONT-y3: The 6G management system should support the ontology to reference and map to existing 3GPP data models (e.g., NRM, performance measurements, alarm definitions).

+REQ-ONT-y2: The 6G management system should support the ontology to reference and map to existing 3GPP data models (e.g., NRM, performance measurements, alarm definitions).

* * * Next change * * * *

S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online1.docx -> 263545_final_S5-263545 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

--- S5-263545d6 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues-20260828_online1.docx

+++ 263545_final_S5-263545 (revision of S5-262928) KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues.docx

@@ -2,3 +2,3 @@

Prague, Czech Republic, 24 – 28 Aug 2026Revision of S5-262928

-Source:Ericsson (moderator)

+Source:Moderator (Ericsson), China Mobile, China Unicom, Ericsson

Title:KSM1- Rel-20 pCR TR32.801-01 Solution on knowledge ontology and other key issues

S5-263546 - KSM2：质量、验证、生成、冲突与溯源

S5-262929 Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx -> S5-263546d1 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

--- S5-262929 Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

+++ S5-263546d1 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

@@ -102,5 +102,5 @@

In some scenarios, it may be beneficial for a group of knowledge MnS producers to have access to a common knowledge instance (NOTE). For example, knowledge MnS producers, each providing knowledge related to a different feature (e.g., energy saving, performance optimization) can have access to a common knowledge instance related to managed element, so they can collaborate to provide a more comprehensive knowledge related to that managed element.

-NOTE: A knowledge instance is a single unit/portion of knowledge that is individually identifiable within the 6G management system. A knowledge instance has its own lifecycle, can be shared, can be versioned, and consumed by different consumers. In ontology terms, the knowledge instance concept maps to the definition of ABox (Assertional Box, i.e., the set of individual assertions in an ontology [x1]).

+NOTE: A knowledge instance is a a uniquely identifiable (set of) portion(s) of knowledge within the 6G management system at a given point in time, that can be individually referenced, exchanged, or consumed..

Knowledge sharing is a feature that allows a group of knowledge MnS producers to have access to one or more knowledge instances, these referred to as shared knowledge instances. The producers can contribute to (and consume from) these knowledge instances. Knowledge sharing benefits from decoupling the knowledge instances from the knowledge MnS producers. The decoupling enables that in case one knowledge MnS producer is not available, the rest of producers of the group can still access the shared knowledge instances.

-Editor’s Note: since the definition of knowledge instance applies to features other than knowledge sharing, the definition may need to be eventually need to move to another clause. Which clause is FFS.

+Editor’s Note: since the definition of knowledge instance applies to features other than knowledge sharing, the definition may need to be eventually need to be move to another clause. Which clause is FFS.

7.2.6.2Potential Requirements

@@ -123,12 +123,11 @@

7.2.x.1Description

-In 6G management system, knowledge may enter the knowledge base from heterogenous sources and through different pathways, and each pathway introduces potential errors or semantic drift.

-Knowledge consumers may use knowledge for decision-making and/or creating new knowledge. The knowledge consumers need assurance that the accessed knowledge is correct and consistent. If the knowledge consumer has no means to determine whether the knowledge has been validated, it must either blindly trust all available knowledge (risking decisions based on incorrect or stale content) or reject all knowledge that lacks verification evidence (losing access to valid information). Without validation, a knowledge base degrades into an unreliable collection of assertions that cannot be trusted by knowledge consumers.

-Validation cannot be a one-time event. Knowledge that was valid at the time of creation may become invalid due to changes in network state, topology reconfigurations, or context evolution. Previously consistent assertions may become contradictory when an event alters the conditions under which they were produced. Therefore, 6G management system needs to support validation not only before knowledge first enters into the knowledge base, but also throughout its lifecycle as conditions evolve.

+The knowledge consumers need assurance that the knowledge they access is valid. A valid knowledge is a knowledge that it is conformant to is conformant to the declared schema, and is semantically correct (e.g., knowledge representing a relationship between two entities is semantically permissible given their types). A knowledge that is made available for consumption through a MnS interface needs to be valid.

+Validation cannot be a one-time event. Knowledge that was valid at the time of creation may become invalid due to changes in network state, topology reconfigurations, or context evolution. Therefore, 6G management system needs to support validation not only before knowledge first gets created, but also throughout its lifecycle as conditions evolve.

This key issue focuses on the following aspects:

When validation occurs, considering that knowledge may need to be validated at multiple points, not only upon initial creation.

-What validation checks are applied to knowledge, and whether different types of knowledge require different validation approaches.

-How information related to knowledge validation (e.g., when/what validation decision was done, by whom, etc) is recorded, enabling post-hoc determination of whether validation was correctly performed.

+What validation checks are applied to knowledge.

+How information related to knowledge validation is recorded.

7.2.x.2Potential Requirements

REQ-KNOWLEDGE-VALIDATION-01: The 6G management system should support validating knowledge instances through their lifecycle.

-REQ-KNOWLEDGE-VALIDATION-02: The 6G management system should support recording information related to knowledge validation to enable traceability and accountability.

+REQ-KNOWLEDGE-VALIDATION-02: The 6G management system should support recording information related to knowledge validation

* * * Next Change * * * *

@@ -136,9 +135,10 @@

7.2.x.1Description

-Knowledge generation refers to the construction of new knowledge instances that did not previously exist in the knowledge base, as opposed to the ingestion or registration of pre-existing facts. Generated knowledge is produced by applying different methods (e.g., statistical inference, rule-based synthesis, aggregation) over data and/or existing knowledge instances. When knowledge is generated using data, the entity creating the knowledge may become a consumer of a shared data framework, e.g., Data Management Framework (DMFW). When knowledge is generated using knowledge instances, these instances belonging to the same or different domains, the entity creating the knowledge becomes knowledge MnS consumer.

-Generated knowledge may be of different types depending on what it represents. For example, causal knowledge captures cause-effect relationships between network entities (e.g., "high load on cell X causes increased handover failure to cell Y"). Similarly, descriptive knowledge captures characterizations of observed patterns or behavior (e.g., "cell X exhibits a recurrent traffic peak between 15:00 and 17:00 on weekdays"). Different knowledge types may require different generation approaches and carry different temporal validity characteristics.

-In 6G management system, different entities can generate knowledge. Knowledge that is generated and exposed to authorized consumers need to be assessable with respect to quality such as its origin, confidence, freshness and scope, so that the consumer can make informed decisions about whether to rely on it. Additionally, knowledge generated from volatile data may become deprecated as network conditions evolve; without mechanisms to track the confidence and freshness of generated knowledge, deprecated knowledge can silently degrade decisions made by knowledge consumers and degrade quality of another knowledge that used it as a source.

+Knowledge generation refers to the creation of new knowledge instances that did not previously exist in a knowledge base. Generated knowledge is produced by applying different methods (e.g., statistical inference, rule-based synthesis, aggregation) over data and/or existing knowledge instances.

+NOTE: Knowledge generation covers both creation from raw data where no prior knowledge existed, and derivation of new knowledge instances from existing ones. In both cases, the result is a new, referenceable knowledge instance that was not previously present in the knowledge base.

+Generated knowledge may be of different types depending on what it represents. For example, causal knowledge captures cause-effect relationships between network entities (e.g., "high load on cell X causes increased handover failure to cell Y"). Similarly, descriptive knowledge captures characterizations of observed patterns or behavior (e.g., "cell X exhibits a recurrent traffic peak between 15:00 and 17:00 on weekdays"). Different knowledge types may require different generation approaches and carry different validity characteristics.

+In 6G management system, different entities can generate knowledge, based on multiple triggers. When knowledge is generated, information associated to its generation is recorded. As the network conditions evolve or time passes, the validity and quality (e.g., freshness, confidence) of that knowledge may change, that change being influenced by the generation method used and the volatility of the underlying data/knowledge from which it was derived.

This key issue focuses on the following aspects:

-What triggers initiate knowledge generation.

+What conditions initiate knowledge generation, without prescribing an exhaustive enumeration of all possible triggers.

What provenance information is recorded for knowledge at generation time.

-How to manage the validity of a generated knowledge, including criteria and conditions for re-generation when knowledge freshness and confidence degrade.

+How to manage the validity of a generated knowledge, as network conditions evolve or time passes.

Whether and how knowledge generation operates in batch mode or incremental mode, and the implications of each mode on consistency and freshness.

@@ -148,3 +148,2 @@

REQ-KNOWLEDGE-GENERATION-03: The 6G management system should support managing the validity of generated knowledge.

-REQ-KNOWLEDGE-GENERATION-04: The 6G management system should support knowledge generation in batch mode and/or incremental mode, with configurable selection of mode.

* * * Next Change * * * *

@@ -152,5 +151,5 @@

7.2.x.1Description

-A knowledge conflict occurs when two or more assertions within a knowledge instance are mutually inconsistent under the governing ontology or domain constraints. Each conflicting assertion may be individually valid within its originating context; however, the conflict emerges only when the assertions are present in the same knowledge instance. Conflicts are different from validation failures: validation rejects invalid content, while conflict handling resolves contradictions between individually valid assertions. The likelihood of conflicts increases in knowledge sharing scenarios (key issue #6), where multiple knowledge MnS producers contribute to a common knowledge instance.

-In the 6G management system, conflicts related to knowledge may differ in nature. Some conflicts arise from mutually exclusive assertions about the same entity at the same time (e.g., "network slice 1 is capacity-constrained at time T1" and "network slice 1 has spare capacity at time T1" are factual contradictions). Other conflicts stem from assertions that are individually valid but apply under different temporal or operational contexts (e.g., "cell X has 40% remaining power budget (daytime)" and "cell Y has 10% remaining power budget (night-time, after high load)" are both factually correct but appear contradictory if the context is omitted). Additionally, knowledge conflicts may emerge when different knowledge producers generate non-compatible assertions for the same resource (e.g., an AI/ML inference function asserts "anomaly detected o cell X", while a rule-based analyser asserts "cell X behaviour is normal, both operating on the same data but reaching contradictory conclusions). Different conflict natures may require different handling approaches.

-Conflicts do not emerge exclusively at one moment in the knowledge lifecycle. They may arise at contribution time, but also post-hoc when previously consistent assertions become contradictory due to external context changes.

+A knowledge conflict occurs when two or more assertions within a knowledge become mutually exclusive. Each conflicting assertion may be individually valid within its originating context; however, the conflict emerges only when these assertions are combined into a common knowledge state. Conflicts are different from validation failures: validation rejects invalid content, while conflict handling resolves contradictions between individually valid assertions.

+In the 6G management system, conflicts related to knowledge may differ in nature. Some conflicts arise from mutually exclusive assertions about the same entity at the same time (e.g., "cell X isServedBy gNB-1" and "cell X isServedBy gNB-2", where the schema constrains a cell to be served by exactly one gNB at a time). Other knowledge conflicts arise when different knowledge producers generate non-compatible assertions for the same resource (e.g., an AI/ML inference function asserts "anomaly detected o cell X", while a rule-based analyser asserts "cell X behaviour is normal, both operating on the same data but reaching contradictory conclusions). Different conflict natures may require different handling approaches.

+Conflicts do not emerge exclusively at one moment in the knowledge lifecycle. They may arise at contribution time, but also post-hoc when previously consistent assertions become contradictory due to e.g., external context changes.

This key issue focuses on the following aspects:

@@ -158,3 +157,2 @@

How the system handles conflicts that cannot be resolved.

-Whether and how to prevent inconsistent information from being preserved in the knowledge base.

7.2.x.2Potential Requirements

@@ -162,3 +160,2 @@

REQ-KNOWLEDGE-CONFLICT-02: The 6G management system should support handling of non-resolved conflicts related to knowledge.

-REQ-KNOWLEDGE-CONFLICT-03: The 6G management system should prevent inconsistencies in the knowledge base.

* * * Next Change * * * *

@@ -166,12 +163,11 @@

7.2.x.1Description

-Knowledge instances in the 6G management system may be derived from other knowledge instances, produced from data, or result from a chain of transformations. As the knowledge base grows, knowledge instances become interconnected through derivation relationships. An operator investigating a faulty autonomous action needs to trace backward from the decision to the knowledge that informed it, then from that knowledge to the data and reasoning that produced it. Likewise, a regulatory audit may require demonstrating the full chain from raw observation to operational knowledge.

-The 6G management system needs to be able to trace the origin, derivation path, and transformation history of every knowledge instance it holds. Every knowledge instance carries a provenance record (see key issue: knowledge generation) that includes references to parent knowledge instances from which it was derived. These parent references, when linked across multiple knowledge instances, form derivation chains that enable traceability.

-The 6G management system should maintain lineage information for every knowledge instance, enabling traceability across the lineage chain, including forward traceability (from source to derived knowledge), backward traceability (from knowledge to its origins), and impact analysis (from a change in any node in the lineage chain to all affected downstream knowledge).

+A knowledge instance in the 6G management system may be derived from different data and other knowledge instances, As knowledge base grows, knowledge instances become interconnected through derivation relationships

+Knowledge lineage is the chain of derivation relationships for any knowledge instance, such that the system can determine what it was derived from, and what depends on it.

+The 6G management system should maintain lineage information enabling traceability across the derivation chain, including forward traceability (from source to derived knowledge), backward traceability (from knowledge to its origins), and impact analysis (from a change in any node in the lineage chain to all affected downstream knowledge).

This key issue focuses on the following aspects:

-How derivation chains across multiple knowledge instances are modelled and stored.

-How the system supports transversing the lineage chain in both directions: from source to all derived knowledge, and from a knowledge instance back to its sources.

-How lineage information enables determining which knowledge instances are affected by changes to upstream knowledge instance or source data (see NOTE)

-NOTE: Changes on upstream knowledge may include invalidation, confidence score, among others.

+How derivation chains between knowledge instances are modelled and stored to form lineage chains.

+How the system supports transversing the lineage chain in both directions: forward (from source to all derived knowledge), and backwards (from a knowledge instance back to its sources).

+How lineage information enables determining which knowledge instances are affected by changes to upstream knowledge instance or source data.

7.2.x.2Potential Requirements

-REQ-KNOWLEDGE-LINEAGE-01: The 6G management system should support linking knowledge instances through their provenance records to form traversable derivation chains.

+REQ-KNOWLEDGE-LINEAGE-01: The 6G management system should support recording derivation relationships between knowledge instances to form traversable lineage chains.

REQ-KNOWLEDGE-LINEAGE-02: The 6G management system should support forward and backward transversal of knowledge lineage chains.

S5-263546d1 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx -> S5-263546d2 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

--- S5-263546d1 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

+++ S5-263546d2 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

@@ -92,8 +92,7 @@

How to apply common data management terminologies, common data, data-set or knowledge properties and representations.

-4.How knowledge is represented by graphs. This includes schema definition of knowledge graphs (node types, edge relationships, attribute), update mechanisms and knowledge graph types (e.g., spatial knowledge graph to represent the quantitative correlations among network metrics within a time window, spatiotemporal knowledge graph to represent the temporal evolution of network metric correlations).

-5.How to ensure the quality of knowledge. This includes definition of quality (e.g., definitions of quality metrics such as semantic accuracy, timeliness, completeness, and trustworthiness), quality evaluation and measurement.

+4.How knowledge is represented by graphs. This includes schema definition of knowledge graphs (node types, edge relationships, attribute).

+5.

Editor’ s Note: Addressing the integrity of Knowledge is FFS.

7.2.1.2Potential Requirements

-PR-KNOWLEDGE-REPRESENTAION-01: The 3GPP management system should support knowledge creation based on management data or existing knowledge.

-PR-KNOWLEDGE-REPRESENTAION-02: The 3GPP management system should support representing knowledge using knowledge graphs.

+PR-KNOWLEDGE-REPRESENTAION-01: The 3GPP management system should support representing knowledge using knowledge graphs.

* * * Next Change * * * *

@@ -102,3 +101,3 @@

In some scenarios, it may be beneficial for a group of knowledge MnS producers to have access to a common knowledge instance (NOTE). For example, knowledge MnS producers, each providing knowledge related to a different feature (e.g., energy saving, performance optimization) can have access to a common knowledge instance related to managed element, so they can collaborate to provide a more comprehensive knowledge related to that managed element.

-NOTE: A knowledge instance is a a uniquely identifiable (set of) portion(s) of knowledge within the 6G management system at a given point in time, that can be individually referenced, exchanged, or consumed..

+NOTE: A knowledge instance is a uniquely identifiable (set of) portion(s) of knowledge within the 6G management system at a given point in time that can be individually exchanged or consumed.

Knowledge sharing is a feature that allows a group of knowledge MnS producers to have access to one or more knowledge instances, these referred to as shared knowledge instances. The producers can contribute to (and consume from) these knowledge instances. Knowledge sharing benefits from decoupling the knowledge instances from the knowledge MnS producers. The decoupling enables that in case one knowledge MnS producer is not available, the rest of producers of the group can still access the shared knowledge instances.

@@ -108,14 +107,16 @@

* * * Next Change * * * *

-7.2.XKey Issue #X: Knowledge Health Profile Management

+7.2.XKey Issue #X: Knowledge Health Information Management

7.2.X.1Description

-Communication network management is increasingly evolving toward automated and autonomous operation, wherein management functions consume information and knowledge obtained from one or more network and management entities. A MnS producer provide management information or knowledge to an MnS consumer for performing functions such as monitoring, analysis, assurance, optimization, configuration, and network control. Such knowledge is observed or retrieved from a source or is generated through aggregation, computation, estimation, inference, prediction, analytics, or artificial-intelligence/machine-learning based processing. Consequently, management knowledge is associated with various characteristics relating to its freshness, origin, temporal state, coverage, completeness.

-In distributed AI environments, each AI agent has certain limitations for knowledge readiness before autonomous decision making. For instance, knowledge can be only partial, knowledge continuously changes, knowledge becomes stale, knowledge may conflict with other sources, knowledge quality varies etc.

-Consequently, using such knowledge, AI agents may be prone to issues such as make incorrect decisions, repeatedly query identical information, may collaborate inefficiently, produce conflicting outputs or violate operational policies etc.

+Communication network management is increasingly evolving toward automated and autonomous operation, wherein management functions consume information and knowledge obtained from one or more network and management entities. A MnS producer provides management information or knowledge to an MnS consumer for performing functions such as monitoring, analysis, assurance, optimization, configuration, and network control. Such knowledge is observed or retrieved from a source or is generated through aggregation, computation, estimation, inference, prediction, analytics, or artificial-intelligence/machine-learning based processing. Consequently, management knowledge is associated with various characteristics relating to its generation, freshness, origin, temporal state, coverage, completeness, quality scope, validity.

+Some of these characteristics does not change throughout the knowledge lifecycle (immutable traits), while others change as time passes, for example, freshness, consistency, provenance, context or network state changes (mutable traits). An MnS consumer that accesses knowledge needs to have visibility on such characteristics associated with knowledge in order to make informed decisions about whether and how to use this knowledge.

+NOTE: These traits result from knowledge management capabilities that include knowledge generation, knowledge quality, validation, etc.

+In distributed AI environments, each producer has certain limitations for knowledge readiness before autonomous decision making. For instance, knowledge can be only partial, knowledge continuously changes, knowledge becomes stale, knowledge may conflict with other sources, knowledge quality varies etc.

+Consequently, using such knowledge, a MnS producer may be prone to issues such as make incorrect decisions, repeatedly query identical information, may collaborate inefficiently, produce conflicting outputs or violate operational policies etc.

By knowing the deficiency arising from such stale, incomplete, or inaccessible network knowledge, actions can be taken that prevents degradation of the quality, correctness, and explainability of AI-driven management and automation decisions.

-Knowledge Health Profile is a dynamic, machine-processable representation of characteristics that are indicative of a quality, or fitness properties of knowledge that is available from, or generated by, a knowledge-providing entity. The characteristics represented in the Knowledge Health Profile includes, temporal characteristics such as freshness, completeness, coverage, provenance, derivation characteristics. An MnS producer characterizes the quality or health of knowledge that it provides by generating or updating a Knowledge Health Profile. A MnS consumer (management task or MnS consuming function) knows the quality of knowledge that it requires. A knowledge debt is derived as a residual gap between the quality of knowledge required by the consumer and the quality of knowledge indicated from the Knowledge Health Profile as provided by the producer.

+Knowledge Health Information is a dynamic, machine-processable representation of characteristics that are indicative of a quality, or fitness properties of knowledge that is available from, or generated by, a knowledge-providing entity. The characteristics represented in the knowledge health information includes, temporal characteristics such as generation, freshness, completeness, coverage, provenance, consistency, validity, derivation characteristics. An MnS producer characterizes the quality or status of knowledge that it provides by generating or updating a Knowledge health information. A MnS consumer (management task or MnS consuming function) knows the adequacy of knowledge that it requires. A knowledge assessment is derived as a residual gap between the quality of knowledge required by the consumer and the quality of knowledge indicated from the Knowledge health information as provided by the producer.

7.2.X.2Potential Requirements

-REQ-KNOWLEDGE-KHP-01: The 6G management system should enable an MnS producer to create and maintain a Knowledge Health Profile.

-REQ-KNOWLEDGE-KHP-02: The 6G management system should enable an authorised MnS consumer to request and receive a Knowledge Health Profile from MnS producer(s).

-REQ-KNOWLEDGE-KHP-03: The 6G management system should enable an MnS producer to dynamically update Knowledge Health Profile in response to changes in knowledge sources, data information, or operational conditions.

-REQ-KNOWLEDGE-KHP-04: The 6G management system should enable an MnS producer to notify an authorised MnS consumer of the changes in Knowledge Health Profile.

+REQ-KNOWLEDGE-KHP-01: The 6G management system should enable an MnS producer to create and maintain a Knowledge health information associated with knowledge instances.

+REQ-KNOWLEDGE-KHP-02: The 6G management system should enable an authorised MnS consumer to request and receive a Knowledge health information from MnS producer(s).

+REQ-KNOWLEDGE-KHP-03: The 6G management system should enable an MnS producer to dynamically update Knowledge health information in response to changes in knowledge sources, data information, or operational conditions.

+REQ-KNOWLEDGE-KHP-04: The 6G management system should enable an MnS producer to notify an authorised MnS consumer of the changes in Knowledge health information.

* * * Next Change * * * *

@@ -123,6 +124,10 @@

7.2.x.1Description

-The knowledge consumers need assurance that the knowledge they access is valid. A valid knowledge is a knowledge that it is conformant to is conformant to the declared schema, and is semantically correct (e.g., knowledge representing a relationship between two entities is semantically permissible given their types). A knowledge that is made available for consumption through a MnS interface needs to be valid.

-Validation cannot be a one-time event. Knowledge that was valid at the time of creation may become invalid due to changes in network state, topology reconfigurations, or context evolution. Therefore, 6G management system needs to support validation not only before knowledge first gets created, but also throughout its lifecycle as conditions evolve.

+Knowledge is valid when is conformant to the declared schema, and is semantically correct (e.g., knowledge representing a relationship between two entities is semantically permissible given their types). Only knowledge that is valid is exposed (made available for consumption) through a MnS interface. When 6G management system suspects that a knowledge (portion) is not valid, the system marks it as suspected invalid. , and it is therefore not exposed through an MnS interface.

+Validation cannot be a one-time event. Knowledge that was valid at the time of creation may become invalid due to changes in network state, topology reconfigurations, quality degradation or context evolution. Therefore, 6G management system needs to support validation not only before knowledge first gets created, but also throughout its lifecycle as conditions evolve.

+The outcome of knowledge validation has an impact on the MnS interface in two scenarios:

+When knowledge that was previously consumed while valid is subsequently marked as suspected, the knowledge MnS producer needs to inform the affected knowledge MnS consumer(s). This enables consumers to make informed decisions about whether to continue relying on that knowledge or not.

+When knowledge is contributed via a MnS interface, the knowledge MnS producer needs to provide validation feedback to the contributor, indicating whether the knowledge has been accepted and its validation status (e.g., valid or suspected).

This key issue focuses on the following aspects:

When validation occurs, considering that knowledge may need to be validated at multiple points, not only upon initial creation.

+How to manage the validity of knowledge, as network conditions evolve, time passes or quality degrades.

What validation checks are applied to knowledge.

@@ -131,3 +136,3 @@

REQ-KNOWLEDGE-VALIDATION-01: The 6G management system should support validating knowledge instances through their lifecycle.

-REQ-KNOWLEDGE-VALIDATION-02: The 6G management system should support recording information related to knowledge validation

+REQ-KNOWLEDGE-VALIDATION-02: The 6G management system should support managing the validity of knowledge.

* * * Next Change * * * *

@@ -135,5 +140,5 @@

7.2.x.1Description

-Knowledge generation refers to the creation of new knowledge instances that did not previously exist in a knowledge base. Generated knowledge is produced by applying different methods (e.g., statistical inference, rule-based synthesis, aggregation) over data and/or existing knowledge instances.

+Knowledge generation refers to the creation or derivation of new knowledge instances that did not previously exist in a knowledge base. Generated knowledge is produced by applying different methods (e.g., statistical inference, rule-based synthesis, aggregation, prediction, computation, observation) over data and/or existing knowledge instances.

NOTE: Knowledge generation covers both creation from raw data where no prior knowledge existed, and derivation of new knowledge instances from existing ones. In both cases, the result is a new, referenceable knowledge instance that was not previously present in the knowledge base.

-Generated knowledge may be of different types depending on what it represents. For example, causal knowledge captures cause-effect relationships between network entities (e.g., “high load on cell X causes increased handover failure to cell Y”). Similarly, descriptive knowledge captures characterizations of observed patterns or behavior (e.g., “cell X exhibits a recurrent traffic peak between 15:00 and 17:00 on weekdays”). Different knowledge types may require different generation approaches and carry different validity characteristics.

+Generated knowledge may be of different types depending on what it represents. For example, causal knowledge captures cause-effect relationships between network entities (e.g., “high load on cell X causes increased handover failure to cell Y”). Similarly, descriptive knowledge captures characterizations of observed patterns or behavior (e.g., “cell X exhibits a recurrent traffic peak between 15:00 and 17:00 on weekdays”). Different knowledge types may require different generation approaches and carry different quality and validity characteristics.

In 6G management system, different entities can generate knowledge, based on multiple triggers. When knowledge is generated, information associated to its generation is recorded. As the network conditions evolve or time passes, the validity and quality (e.g., freshness, confidence) of that knowledge may change, that change being influenced by the generation method used and the volatility of the underlying data/knowledge from which it was derived.

@@ -141,4 +146,4 @@

What conditions initiate knowledge generation, without prescribing an exhaustive enumeration of all possible triggers.

-What provenance information is recorded for knowledge at generation time.

-How to manage the validity of a generated knowledge, as network conditions evolve or time passes.

+What information is recorded for knowledge at generation time, including provenance information.

+How to manage the validity and quality of a generated knowledge, as network conditions evolve or time passes.

Whether and how knowledge generation operates in batch mode or incremental mode, and the implications of each mode on consistency and freshness.

@@ -146,3 +151,3 @@

REQ-KNOWLEDGE-GENERATION-01: The 6G management system should support knowledge generation based on multiple triggers.

-REQ-KNOWLEDGE-GENERATION-02: The 6G management system should support recording provenance information regarding generated knowledge.

+REQ-KNOWLEDGE-GENERATION-02: The 6G management system should support recording information regarding generated knowledge.

REQ-KNOWLEDGE-GENERATION-03: The 6G management system should support managing the validity of generated knowledge.

@@ -163,3 +168,3 @@

7.2.x.1Description

-A knowledge instance in the 6G management system may be derived from different data and other knowledge instances, As knowledge base grows, knowledge instances become interconnected through derivation relationships

+A knowledge instance in the 6G management system may be derived from different data and other knowledge instances. As knowledge base grows, knowledge instances become interconnected through derivation relationships

Knowledge lineage is the chain of derivation relationships for any knowledge instance, such that the system can determine what it was derived from, and what depends on it.

S5-263546d2 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx -> S5-263546d4 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

--- S5-263546d2 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

+++ S5-263546d4 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

@@ -93,3 +93,2 @@

4.How knowledge is represented by graphs. This includes schema definition of knowledge graphs (node types, edge relationships, attribute).

-5.

Editor’s Note: Addressing the integrity of Knowledge is FFS.

@@ -107,16 +106,15 @@

* * * Next Change * * * *

+7.2.XKey Issue #X: Knowledge Characteristics Exposure

+7.2.X.1Description

7.2.XKey Issue #X: Knowledge Health Information Management

7.2.X.1Description

-Communication network management is increasingly evolving toward automated and autonomous operation, wherein management functions consume information and knowledge obtained from one or more network and management entities. A

MnS producer provides management information or knowledge to an MnS consumer for performing functions such as monitoring, analysis, assurance, optimization, configuration, and network control. Such knowledge is observed or retrieved from a source or is generated through aggregation, computation, estimation, inference, prediction, analytics, or artificial-intelligence/machine-learning based processing. Consequently, management knowledge is associated with various characteristics relating to its generation, freshness, origin, temporal state, coverage, completeness, quality scope, validity.

-Some of these characteristics does not change throughout the knowledge lifecycle (immutable traits), while others change as time passes, for example, freshness, consistency, provenance, context or network state changes (mutable traits). An MnS consumer that accesses knowledge needs to have visibility on such characteristics associated with knowledge in order to make informed decisions about whether and how to use this knowledge.

-NOTE: These traits result from knowledge management capabilities that include knowledge generation, knowledge quality, validation, etc.

-In distributed AI environments, each producer has certain limitations for knowledge readiness before autonomous decision making. For instance, knowledge can be only partial, knowledge continuously changes, knowledge becomes stale, knowledge may conflict with other sources, knowledge quality varies etc.

-Consequently, using such knowledge, a MnS producer may be prone to issues such as make incorrect decisions, repeatedly query identical information, may collaborate inefficiently, produce conflicting outputs or violate operational policies etc.

-By knowing the deficiency arising from such stale, incomplete, or inaccessible network knowledge, actions can be taken that prevents degradation of the quality, correctness, and explainability of AI-driven management and automation decisions.

-Knowledge Health Information is a dynamic, machine-processable representation of characteristics that are indicative of a quality, or fitness properties of knowledge that is available from, or generated by, a knowledge-providing entity. The characteristics represented in the knowledge health information includes, temporal characteristics such as generation, freshness, completeness, coverage, provenance, consistency, validity, derivation characteristics. An MnS producer characterizes the quality or status of knowledge that it provides by generating or updating a Knowledge health information. A MnS consumer (management task or MnS consuming function) knows the adequacy of knowledge that it requires. A knowledge assessment is derived as a residual gap between the quality of knowledge required by the consumer and the quality of knowledge indicated from the Knowledge health information as provided by the producer.

+Communication network management is increasingly evolving toward automated and autonomous operation, wherein management functions consume information and knowledge obtained from one or more network and management entities. A MnS producer provides management information or knowledge to an MnS consumer for performing functions such as monitoring, analysis, assurance, optimization, configuration, and network control. Such knowledge is observed or retrieved from a source or is generated through aggregation, computation, estimation, inference, prediction, analytics, or artificial-intelligence/machine-learning based processing. Consequently, management knowledge is associated with various characteristics relating to its generation, freshness, origin, temporal state, coverage, completeness, quality, scope, validity etc.

+Some of these characteristics does not change throughout the knowledge lifecycle (immutable traits), while others change as time passes, for example, freshness, consistency, context or network state changes (mutable traits). An MnS consumer that accesses knowledge needs to have visibility on such characteristics associated with knowledge in order to make informed decisions about whether and how to use this knowledge.

+NOTE: These characteristics result from knowledge management capabilities that include knowledge generation, knowledge quality maintenance, validation, etc.

+Knowledge Health Information is a dynamic, machine-processable representation of characteristics that are indicative of a quality, or fitness properties of knowledge that is available from, or generated by, a knowledge-providing entity. The characteristics represented in the knowledge health information includes temporal characteristics such as generation, freshness, completeness, coverage, consistency, validity, derivation characteristics. An MnS producer characterizes the quality or status of knowledge that it provides by generating or updating a knowledge health information. A MnS consumer (management task or MnS consuming function) knows the adequacy of knowledge that it requires. A knowledge assessment is derived as a residual gap between the quality of knowledge required by the consumer and the quality of knowledge indicated from the Knowledge health information as provided by the producer.

+NOTE: The relation between this key issue and Knowledge Validation key issue is to be further evaluated.

7.2.X.2Potential Requirements

-REQ-KNOWLEDGE-KHP-01: The 6G management system should enable an MnS producer to create and maintain a Knowledge health information associated with knowledge instances.

-REQ-KNOWLEDGE-KHP-02: The 6G management system should enable an authorised MnS consumer to request and receive a Knowledge health information from MnS producer(s).

-REQ-KNOWLEDGE-KHP-03: The 6G management system should enable an MnS producer to dynamically update Knowledge health information in response to changes in knowledge sources, data information, or operational conditions.

-REQ-KNOWLEDGE-KHP-04: The 6G management system should enable an MnS producer to notify an authorised MnS consumer of the changes in Knowledge health information.

+REQ-KNOWLEDGE-KHP-01: The 6G management system should enable a knowledge MnS producer to create and maintain knowledge characteristics associated with a knowledge instance.

+REQ-KNOWLEDGE-KHP-02: The 6G management system should enable an authorised MnS consumer to request knowledge characteristics associated with a knowledge instance from MnS producer.

+REQ-KNOWLEDGE-KHP-03: The 6G management system should enable a knowledge MnS producer to provide requested characteristics associated with a knowledge instance to authorized MnS consumer.

* * * Next Change * * * *

@@ -134,2 +132,3 @@

How information related to knowledge validation is recorded.

+Editor’ s note: The relationship between this key issue and “Knowledge Health Information Management” is for further evaluation.

7.2.x.2Potential Requirements

@@ -161,20 +160,5 @@

When and how conflicts are detected in the knowledge instance.

-How the system handles conflicts that cannot be resolved.

+NOTE: How the system handles conflicts that cannot be resolved, and whether this has an impact on the MnS interface, is FFS.

7.2.x.2Potential Requirements

REQ-KNOWLEDGE-CONFLICT-01: The 6G management system should support detection of conflicts related to knowledge.

-REQ-KNOWLEDGE-CONFLICT-02: The 6G management system should support handling of non-resolved conflicts related to knowledge.

-* * * Next Change * * * *

-7.2.xKey Issue #<X>: Knowledge Lineage

-7.2.x.1Description

-A knowledge instance in the 6G management system may be derived from different data and other knowledge instances. As knowledge base grows, knowledge instances become interconnected through derivation relationships

-Knowledge lineage is the chain of derivation relationships for any knowledge instance, such that the system can determine what it was derived from, and what depends on it.

-The 6G management system should maintain lineage information enabling traceability across the derivation chain, including forward traceability (from source to derived knowledge), backward traceability (from knowledge to its origins), and impact analysis (from a change in any node in the lineage chain to all affected downstream knowledge).

-This key issue focuses on the following aspects:

-How derivation chains between knowledge instances are modelled and stored to form lineage chains.

-How the system supports transversing the lineage chain in both directions: forward (from source to all derived knowledge), and backwards (from a knowledge instance back to its sources).

-How lineage information enables determining which knowledge instances are affected by changes to upstream knowledge instance or source data.

-7.2.x.2Potential Requirements

-REQ-KNOWLEDGE-LINEAGE-01: The 6G management system should support recording derivation relationships between knowledge instances to form traversable lineage chains.

-REQ-KNOWLEDGE-LINEAGE-02: The 6G management system should support forward and backward transversal of knowledge lineage chains.

-REQ-KNOWLEDGE-LINEAGE-03: The 6G management system should support determining which knowledge instances are impacted by changes to upstream knowledge instances or source data.

* * * End of Changes * * * *

S5-263546d4 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx -> S5-263546d6 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

--- S5-263546d4 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

+++ S5-263546d6 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

@@ -1,3 +1,3 @@

-3GPP TSG SA5 Meeting #168S5-262929

-Prague, Czech Republic, 24-28 August 2026

+3GPP TSG SA5 Meeting #168S5-263546

+Prague, Czech Republic, 24-28 August 2026Revision of S5-262929

Source:Moderator (China Unicom)

@@ -89,3 +89,3 @@

This key issue focuses on the following aspects:

-1.How knowledge can represent information related to network operation features (e.g., conflict detection, performance management, energy saving, etc), so that knowledge can be used for network operation.

+1.How knowledge can represent information related to network operation features (e.g., conflict detection, performance management, energy saving, etc.), so that knowledge can be used for network operation.

How relationships among management data (e.g., managed entities, configuration data, KPIs, performance metrics, alarms, trace information) can be formalized into knowledge. This includes studying which relationships should be included in knowledge and how the relationships should be represented.

@@ -102,3 +102,3 @@

Knowledge sharing is a feature that allows a group of knowledge MnS producers to have access to one or more knowledge instances, these referred to as shared knowledge instances. The producers can contribute to (and consume from) these knowledge instances. Knowledge sharing benefits from decoupling the knowledge instances from the knowledge MnS producers. The decoupling enables that in case one knowledge MnS producer is not available, the rest of producers of the group can still access the shared knowledge instances.

-Editor’ s Note: since the definition of knowledge instance applies to features other than knowledge sharing, the definition may need to be eventually need to be move to another clause. Which clause is FFS.

+Editor’ s Note: since the definition of knowledge instance applies to features other than knowledge sharing, the definition may need to be eventually need to be moved to another clause. Which clause is FFS.

7.2.6.2Potential Requirements

@@ -122,10 +122,10 @@

7.2.x.1Description

-Knowledge is valid when is conformant to the declared schema, and is semantically correct (e.g., knowledge representing a relationship between two entities is semantically permissible given their types). Only knowledge that is valid is exposed (made available for consumption) through a MnS interface. When 6G management system suspects that a knowledge (portion) is not valid, the system marks it as suspected invalid. , and it is therefore not exposed through an MnS interface.

-Validation cannot be a one-time event. Knowledge that was valid at the time of creation may become invalid due to changes in network state, topology reconfigurations, quality degradation or context evolution. Therefore, 6G management system needs to support validation not only before knowledge first gets created, but also throughout its lifecycle as conditions evolve.

+Knowledge is valid when is conformant to the declared schema, and is semantically correct (e.g., knowledge representing a relationship between two entities is semantically permissible given their types) and is per required quality or acceptable with respect to some of its characteristics. Only knowledge that is valid is exposed (made available for consumption) through a MnS interface along with associated validation information. When 6G management system suspects that a knowledge (portion) is not valid, the system marks it as suspected invalid, and it is therefore not exposed through an MnS interface.

+Validation cannot be a one-time event. Knowledge that was valid at the time of creation may become invalid due to changes in network state, topology reconfigurations, quality degradation, temporal factors, consistency or context evolution. Therefore, 6G management system needs to support validation not only before knowledge first gets created, but also throughout its lifecycle as conditions evolve.

The outcome of knowledge validation has an impact on the MnS interface in two scenarios:

-When knowledge that was previously consumed while valid is subsequently marked as suspected, the knowledge MnS producer needs to inform the affected knowledge MnS consumer(s). This enables consumers to make informed decisions about whether to continue relying on that knowledge or not.

-When knowledge is contributed via a MnS interface, the knowledge MnS producer needs to provide validation feedback to the contributor, indicating whether the knowledge has been accepted and its validation status (e.g., valid or suspected).

+When knowledge that was previously consumed while valid is subsequently marked as suspected invalid, the knowledge MnS producer needs to inform the affected knowledge MnS consumer(s) along with the associated validation information. This enables consumers to make informed decisions about whether to continue relying on that knowledge or not.

+When knowledge is contributed via a knowledge MnS interface, the knowledge MnS producer needs to provide validation feedback to the contributor, indicating whether the knowledge has been accepted and its validation status (e.g., valid or suspected invalid).

This key issue focuses on the following aspects:

When validation occurs, considering that knowledge may need to be validated at multiple points, not only upon initial creation.

-How to manage the validity of knowledge, as network conditions evolve, time passes or quality degrades.

+How to manage the validity of knowledge, as network conditions evolve, time passes or characteristics (including quality) of knowledge changes.

What validation checks are applied to knowledge.

@@ -135,3 +135,3 @@

REQ-KNOWLEDGE-VALIDATION-01: The 6G management system should support validating knowledge instances through their lifecycle.

-REQ-KNOWLEDGE-VALIDATION-02: The 6G management system should support managing the validity of knowledge.

+REQ-KNOWLEDGE-VALIDATION-02: The 6G management system should support managing the validity of knowledge based on certain knowledge characteristics.

* * * Next Change * * * *

@@ -146,3 +146,2 @@

What information is recorded for knowledge at generation time, including provenance information.

-How to manage the validity and quality of a generated knowledge, as network conditions evolve or time passes.

Whether and how knowledge generation operates in batch mode or incremental mode, and the implications of each mode on consistency and freshness.

@@ -151,3 +150,2 @@

REQ-KNOWLEDGE-GENERATION-02: The 6G management system should support recording information regarding generated knowledge.

-REQ-KNOWLEDGE-GENERATION-03: The 6G management system should support managing the validity of generated knowledge.

* * * Next Change * * * *

S5-263546d6 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx -> S5-263546d7 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

--- S5-263546d6 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

+++ S5-263546d7 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

@@ -106,4 +106,2 @@

* * * Next Change * * * *

-7.2.XKey Issue #X: Knowledge Characteristics Exposure

-7.2.X.1Description

7.2.XKey Issue #X: Knowledge Health Information Management

@@ -111,3 +109,3 @@

Communication network management is increasingly evolving toward automated and autonomous operation, wherein management functions consume information and knowledge obtained from one or more network and management entities. A MnS producer provides management information or knowledge to an MnS consumer for performing functions such as monitoring, analysis, assurance, optimization, configuration, and network control. Such knowledge is observed or retrieved from a source or is generated through aggregation, computation, estimation, inference, prediction, analytics, or artificial-intelligence/machine-learning based processing. Consequently, management knowledge is associated with various characteristics relating to its generation, freshness, origin, temporal state, coverage, completeness, quality, scope, validity etc.

-Some of these characteristics does not change throughout the knowledge lifecycle (immutable traits), while others change as time passes, for example, freshness, consistency, context or network state changes (mutable traits). An MnS consumer that accesses knowledge needs to have visibility on such characteristics associated with knowledge in order to make informed decisions about whether and how to use this knowledge.

+Some of these characteristics does not change throughout the knowledge lifecycle (immutable characteristics), while others change as time passes, for example, freshness, consistency, context or network state changes (mutable characteristics). An MnS consumer that accesses knowledge needs to have visibility on such characteristics associated with knowledge in order to make informed decisions about whether and how to use this knowledge.

NOTE: These characteristics result from knowledge management capabilities that include knowledge generation, knowledge quality maintenance, validation, etc.

@@ -129,3 +127,3 @@

When validation occurs, considering that knowledge may need to be validated at multiple points, not only upon initial creation.

-How to manage the validity of knowledge, as network conditions evolve, time passes or characteristics (including quality) of knowledge changes.

+How to manage the validity and quality of knowledge, as network conditions evolve, time passes or characteristics of knowledge changes including its quality.

What validation checks are applied to knowledge.

@@ -142,6 +140,6 @@

Generated knowledge may be of different types depending on what it represents. For example, causal knowledge captures cause-effect relationships between network entities (e.g., "high load on cell X causes increased handover failure to cell Y"). Similarly, descriptive knowledge captures characterizations of observed patterns or behavior (e.g., "cell X exhibits a recurrent traffic peak between 15:00 and 17:00 on weekdays"). Different knowledge types may require different generation approaches and carry different quality and validity characteristics.

-In 6G management system, different entities can generate knowledge, based on multiple triggers. When knowledge is generated, information associated to its generation is recorded. As the network conditions evolve or time passes, the validity and quality (e.g., freshness, confidence) of that knowledge may change, that change being influenced by the generation method used and the volatility of the underlying data/knowledge from which it was derived.

+In 6G management system, different entities can generate knowledge, based on multiple triggers. When knowledge is generated, information associated to its generation is recorded. As the network conditions evolve or time passes, the validity and quality (e.g., freshness, confidence, completeness) of that knowledge may change, that change being influenced by the generation method used and the volatility of the underlying data/knowledge from which it was derived.

This key issue focuses on the following aspects:

What conditions initiate knowledge generation, without prescribing an exhaustive enumeration of all possible triggers.

-What information is recorded for knowledge at generation time, including provenance information.

+What information is configured and recorded for knowledge at generation time, including provenance information.

Whether and how knowledge generation operates in batch mode or incremental mode, and the implications of each mode on consistency and freshness.

@@ -149,3 +147,3 @@

REQ-KNOWLEDGE-GENERATION-01: The 6G management system should support knowledge generation based on multiple triggers.

-REQ-KNOWLEDGE-GENERATION-02: The 6G management system should support recording information regarding generated knowledge.

+REQ-KNOWLEDGE-GENERATION-02: The 6G management system should support recording and configuring information regarding generated knowledge.

* * * Next Change * * * *

S5-263546d7 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx -> 263546_final_S5-263546 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

--- S5-263546d7 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

+++ 263546_final_S5-263546 (revision of S5-263929) Rel-20 pCR TR 32.801-01 KSM2-New key issues.docx

@@ -2,3 +2,3 @@

Prague, Czech Republic, 24-28 August 2026Revision of S5-262929

-Source:Moderator (China Unicom)

+Source:Moderator (China Unicom), Ericsson, Samsung

Title:KSM2-New key issues

@@ -156,3 +156,3 @@

When and how conflicts are detected in the knowledge instance.

-NOTE: How the system handles conflicts that cannot be resolved, and whether this has an impact on the MnS interface, is FFS.

+Editor’ s note: How the system handles conflicts that cannot be resolved, and whether this has an impact on the MnS interface, is FFS.

7.2.x.2Potential Requirements

S5-263547 - KSM3：架构与建模

S5-262930 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d1 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-262930 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d1 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -47,37 +47,18 @@

8.XKnowledge Management

-8.1Potential Solution #1: Knowledge Management deployment options

+8.X.1Key Issue #2: Management Architecture Aspects of Knowledge

+8.X.1.1Potential Solution #1: Knowledge Management deployment options

The following figure depicts the overall set of entities that could be involved in Knowledge collection, generation and transfer:

Figure 8.1-1: Knowledge Management deployment options: including where i) 3GPP system provides data and Knowledge production happens in management systems; and ii) Knowledge is produced in 3GPP system and provided to management systems

-Management Data Generator: This entity refers to any managed function as defined in 3GPP TS 28.541. This entity generates management data as specified in 3GPP TS 28.552.

-SNM MnS Producer: This entity is responsible to generate Knowledge based on the management data received from the management data generator. This entity is also responsible to maintain the Knowledge Base containing the historical knowledge. When the knowledge producer is part of the 3GPP systems, the management data is augmented by the Knowledge and becomes the part of management data only. E.g., in case of PM, the template defined in 28.552 is extended to include Knowledge as part of performance data.

-SNM MnS Consumer: This entity is responsible for taking network management decision based on the available knowledge. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

-When the knowledge producer is part of the 3GPP systems, Knowledge is managed as part of NRM, otherwise when the knowledge producer is part of the 3GPP management systems, Knowledge is managed as part of Management Data.

-8.X.1Potential Solution #1: Architectural aspects of Knowledge Management

+Knowledge MnS Producer: This entity is responsible to generate Knowledge based on the management data received from the management data generator. This entity is also responsible to maintain the Knowledge Base containing the historical knowledge. When the knowledge producer is part of the 3GPP systems, the management data is augmented by the Knowledge and becomes the part of management data only. E.g., in case of PM, the template defined in 28.552 is extended to include Knowledge as part of performance data.

+Knowledge MnS Consumer: This entity is responsible for taking network management decision based on the available knowledge. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

+When the knowledge producer is part of the 3GPP systems, Knowledge is managed as part of Management Data, otherwise when the knowledge producer is part of the 3GPP management systems, Knowledge is managed as part of NRM.

+8.X.1.2Potential Solution #1: Architectural aspects of Knowledge Management

The following is proposed to be introduced in SA5 to support Knowledge Management and Knowledge-based Management

-Knowledge may be produced and provided by any network node, network function or management function

-Introduce an IOC representing the knowledge production capability, say called KnowledgeFunction. The IOC is name contained on subnetwork or managed element.

-Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true.

-Introduce an IOC representing the MnS consumer's configuration to the MnS producer to start producing knowledge. The production should be a job that allows for single shot or continuous delivery of knowledge. The IOC may be called KnowledgeJob to be name-contained in a KnowledgeFunction. The KnowledgeFunction may contain one or more KnowledgeJob(s).

-The NRM fragments can be represented by Figures Figure 8.X.1-1: and Figure 8.X.1-2:

-@startuml

-skinparam ClassStereotypeFontStyle normal

-skinparam ClassBackgroundColor White

-skinparam shadowing false

-skinparam monochrome true

-skinparam linetype ortho

-hide members
-hide circle
-class ManagedEntity <<ProxyClass>>
-class KnowledgeFunction <<InformationObjectClass>>
-class KnowledgeJob << InformationObjectClass >>
-ManagedEntity "1" *-- "*" KnowledgeFunction: <<names>>
-KnowledgeFunction "1" *-r- "*" KnowledgeJob
-note right of ManagedEntity
-Represents the following IOCs:
-SubNetwork or
-ManagedElement
-end note
-@enduml
-Figure 8.X.1-1: Relations for Data Framework Functionalities
-The KnowledgeFunction MOI can be configured, e.g. by the operator to specify knowledge access rights for different potential consumers
+1)Knowledge may be produced and provided by any management function
+Introduce an object (e.g. an IOC) representing the knowledge production capability, say called KnowledgeFunction..
+Editor's Note: Whether Knowledge can be produced and provided by network nodes and network functions is FFS
+2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true.
+Introduce a KnowledgeJob as an IOC representing the MnS consumer's configuration to the MnS producer to start producing knowledge. The production should be a job that allows for single shot or continuous delivery of knowledge. The KnowledgeJob is associated with the KnowledgeFunction. The KnowledgeFunction may be associated with one or more KnowledgeJob(s).
+3)
+The knowledge production capability (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify knowledge access rights for different potential consumers
Introduce a set of configurable attributes on the KnowledgeFunction. Example configurable attributes include:
@@ -85,3 +66,3 @@
Note: the list of configurable attributes of the KnowledgeFunction can be extended
-The KnowledgeJob can be customized to indicate specific characteristics or properties of the knowledge required by the consumer.
+4)The KnowledgeJob can be customized to indicate specific characteristics or properties of the knowledge required by the consumer.
Introduce a set of attributes on the KnowledgeJob that enable the consumer to scope the required knowledge. Example attributes include:
@@ -89,54 +70,21 @@
Editor's note: the lust of attributes of the KnowledgeJob can be extended
-8.X.APotential Solution #1: Knowledge Representation
+8.X.2Key Issue #1: Knowledge Representation and Management
+8.X.2.1Potential Solution #1: Knowledge Representation
The following is proposed to be introduced in SA5 to support Knowledge representation.
-Knowledge can be contained by a network node, network function, management function or in a report exchanged among entities.
-Introduce an IOC representing the knowledge, say called Knowledge that is name-contained on subnetwork or managed element or a KnowledgeReport.
-Knowledge includes at least one instance of data and the relation of that data to at least one other instance of data. It may as such be an aggregation of two instances of knowledge. The associations between the data or knowledge instances is an explicit relationship that characterizes that association.
-Introduce an IOC representing relationship for a given association between data or knowledge.
-The NRM fragments can be represented by Figure 8.X.A-1:
-@startuml
-skinparam ClassStereotypeFontStyle normal
-skinparam ClassBackgroundColor White
-skinparam shadowing false

-skinparam monochrome true

-'skinparam linetype ortho

-hide members

-hide circle

-class ManagedEntity <<ProxyClass>>

-class Knowledge <<InformationObjectClass>>

-class KnowledgeNode << InformationObjectClass >>

-class KnowledgeRelation << InformationObjectClass >>

-ManagedEntity "1" *-- "*" Knowledge: <<names>>

-Knowledge "1" *-- "*" KnowledgeNode

-Knowledge "1" *-- "*" KnowledgeRelation

-KnowledgeNode "1" -r- "*" KnowledgeRelation

-KnowledgeNode "1" -- "*" KnowledgeNode

-note left of ManagedEntity

-Represents the following IOCs:

-SubNetwork or

-ManagedElement

-KnowledgeReport

-end note

-note left of KnowledgeNode

-Represents one of:

-data or

-Knowledge

-end note

-@enduml

-Figure 8.X.A-1: Knowledge representation

-8.xPotential Solution #x: Knowledge Representation

-Figure 8.1.1-1: Knowledge Representation

-Knowledge is an analytical derivation from data that is natively available for consumer to use for intelligent network management. The Knowledge Base is a collection of available Knowledge. It can be defined using the flowing attributes:

-totalKnowledge: This specifies the total number of available Knowledge

-maxKnowledge: This specifies the total number of Knowledge instance that can be maintained.

-retentionPolicy: This specifies the retention policy of the constituent Knowledge instances. It is defined in terms of hours. The Knowledge older than the defined hours will be deleted.

-Individual Knowledge can be defined using the following common attributes:

-kName: This specifies the name of the Knowledge.

-kUsedPMData: This defines the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

-kGenerationTime: This defines the time stamp at which this knowledge was generated.

-kContext: This defines the context of knowledge. This will contain the fowling information.

-Location: This defines the geographical area to which the knowledge would apply.

-Time: This defines the time at which the knowledge would apply.

-Purpose: This defines the network problem categories where this knowledge can be applied. This will a ENUM representing the MDAType defined in 3GPP TS 28.104. Alternatively, it can be defined a ENUM with probable value of COVERAGE, MOBILITY, SOFTWARE_UPGRADE, RESOURCE_DEPLITION.

-8.x.yPotential Solution #<y>: Knowledge and data relationship

+1)Knowledge can be delivered to the MnS consumer, e.g., via a report.

+Introduce an IOC or dataType representing the knowledge, say called Knowledge. The Knowledge is associated with a KnowledgeFunction, a KnowledgeJob .

+Editor's Note: Whether Knowledge should be an IC or datatype is FFS

+Editor's Note: Whether Knowledge is associated with a KnowledgeReport is FFS

+2)Knowledge indicates a graph of relationships between entities.

+Knowledge should include the entities that are being related as vertices of the graph. Introduce an attribute on Knowledge to represent the vertex, e.g., to be called KnowledgeNode. The KnowledgeNode may for example be instance of data, event, etc. Example content for the KnowledgeNode could be:

+-UsedPMData: which indicates the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

+Knowledge should include the relationships among the entities. Introduce an attribute on Knowledge to represent the relations, e.g., to be called KnowledgeRelation. Example relationship may include "isCausedBy", "isCauseOf", "leadTo", "canResultIn", etc.

+3)Knowledge can be associated with specific context that helps in its interpretability. Introduce an attribute for context, to be called KnowledgeContext. The KnowledgeContext may for example include:

+-Location: This defines the geographical area to which the knowledge would apply.

+-Time: This defines the time e.g., at which the knowledge was generated or would apply.

+-Purpose: This defines the network problem categories where this knowledge can be applied. Example purposes may include: COVERAGE, MOBILITY, SOFTWARE_UPGRADE, RESOURCE_DEPLITION.

+Editor's Note: Whether specific values for Purpose should specified e.g., as ENUM is FFS

+-retentionPolicy: This specifies the retention policy of the constituent Knowledge instances. It is defined in terms of hours. The Knowledge older than the defined hours will be deleted. It may indicate the time for which the MnS consumer is recommended to consider the knowledge as useful beyond which time the knowledge can be deleted.

+8.X.3Key Issue #7: Knowledge Lifecycle Management

+8.X.3.1Potential Solution 1: Knowledge and data relationship

Key issue #7 for knowledge management is Knowledge Lifecycle Management. This key issue, specified in clause 7.2.7 of the present TR, includes two aspects: 1) the lifecycle management of knowledge; and 2) how the lifecycle management of knowledge can be decoupled from the data lifecycle.

S5-263547d1 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d3 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d1 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d3 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -47,6 +47,6 @@

8.XKnowledge Management

-8.X.1Key Issue #2: Management Architecture Aspects of Knowledge

-8.X.1.1Potential Solution #1: Knowledge Management deployment options

+8.X.bPotential Solutions for Key Issue #2: Management Architecture Aspects of Knowledge

+8.X.b.1Potential Solution #1: Knowledge Management deployment options

The following figure depicts the overall set of entities that could be involved in Knowledge collection, generation and transfer:

-Figure 8.1-1: Knowledge Management deployment options: including where i) 3GPP system provides data and Knowledge production happens in management systems; and ii) Knowledge is produced in 3GPP system and provided to management systems

+Figure 8.1-1: Knowledge Management deployment options: including where i) 3GPP system provides data and Knowledge production happens in management systems.

Knowledge MnS Producer: This entity is responsible to generate Knowledge based on the management data received from the management data generator. This entity is also responsible to maintain the Knowledge Base containing the historical knowledge. When the knowledge producer is part of the 3GPP systems, the management data is augmented by the Knowledge and becomes the part of management data only. E.g., in case of PM, the template defined in 28.552 is extended to include Knowledge as part of performance data.

@@ -54,11 +54,12 @@

When the knowledge producer is part of the 3GPP systems, Knowledge is managed as part of Management Data, otherwise when the knowledge producer is part of the 3GPP management systems, Knowledge is managed as part of NRM.

-8.X.1.2Potential Solution #1: Architectural aspects of Knowledge Management

+8.X.b.2Potential Solution #1: Architectural aspects of Knowledge Management

The following is proposed to be introduced in SA5 to support Knowledge Management and Knowledge-based Management

1)Knowledge may be produced and provided by any management function

-Introduce an object (e.g. an IOC) representing the knowledge production capability, say called KnowledgeFunction..

+Introduce an object representing the capability to provide knowledge-related services, say called KnowledgeFunction.

+Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope.

Editor's Note: Whether Knowledge can be produced and provided by network nodes and network functions is FFS

-2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true.

-Introduce a KnowledgeJob as an IOC representing the MnS consumer's configuration to the MnS producer to start producing knowledge. The production should be a job that allows for single shot or continuous delivery of knowledge. The KnowledgeJob is associated with the KnowledgeFunction. The KnowledgeFunction may be associated with one or more KnowledgeJob(s).

+2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

+Introduce a KnowledgeJob to represent the process including the MnS consumer's configuration to the MnS producer for producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

3)

-The knowledge production capability (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify knowledge access rights for different potential consumers

+The knowledge production object (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify the scope for which knowledge can be provided

Introduce a set of configurable attributes on the KnowledgeFunction. Example configurable attributes include:

@@ -69,14 +70,19 @@

-The network scope for which the consumer requires Knowledge to be provided by the MnS producer

-Editor's note: the lust of attributes of the KnowledgeJob can be extended

-8.X.2Key Issue #1: Knowledge Representation and Management

-8.X.2.1Potential Solution #1: Knowledge Representation

+Editor's note: the list of attributes of the KnowledgeJob can be extended

+Editor's note: Whether the KnowledgeJob should allow the MnS consumer to provide knowledge to the MnS producer is FFS.

+8.X.aPotential Solutions for Key Issue #1: Knowledge Representation and Management

+8.X.a.1Potential Solution #1: Knowledge Representation

The following is proposed to be introduced in SA5 to support Knowledge representation.

1)Knowledge can be delivered to the MnS consumer, e.g., via a report.

-Introduce an IOC or dataType representing the knowledge, say called Knowledge. The Knowledge is associated with a KnowledgeFunction, a KnowledgeJob .

-Editor's Note: Whether Knowledge should be an IC or datatype is FFS

-Editor's Note: Whether Knowledge is associated with a KnowledgeReport is FFS

-2)Knowledge indicates a graph of relationships between entities.

-Knowledge should include the entities that are being related as vertices of the graph. Introduce an attribute on Knowledge to represent the vertex, e.g., to be called KnowledgeNode. The KnowledgeNode may for example be instance of data, event, etc. Example content for the KnowledgeNode could be:

--UsedPMData: which indicates the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

-Knowledge should include the relationships among the entities. Introduce an attribute on Knowledge to represent the relations, e.g., to be called KnowledgeRelation. Example relationship may include "isCausedBy", "isCauseOf", "leadTo", "canResultIn", etc.

+Introduce an object representing the knowledge, say called Knowledge. The Knowledge is associated with a KnowledgeFunction, a KnowledgeJob .

+Editor's Note: The nature of the object Knowledge is FFS

+2)Knowledge indicates a graph of relationships between entities. The content to be included as entities or relationships is derived form an ontology.

+Introduce an ontology in SA5 that contains all vocabulary that can be used aspart of entities or relationships in Knowledge.

+Example content for the entities in the ontology could be:

+-content from the 3GPP management system NRM

+-PMData: which indicates the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

+Example relationship in the ontology may include "isCausedBy", "isCauseOf", "leadTo", "canResultIn", etc.

+Editor's note: The full list of items to be included in the vocabulary in the ontology are FFS.

+Knowledge should include the entities that are being related as vertices of the graph. Introduce an attribute on Knowledge to represent the vertex, e.g., to be called KnowledgeNode. The KnowledgeNode may for example be instance of data, event, etc.

+Knowledge should include the relationships among the entities. Introduce an attribute on Knowledge to represent the relations, e.g., to be called KnowledgeRelation.

3)Knowledge can be associated with specific context that helps in its interpretability. Introduce an attribute for context, to be called KnowledgeContext. The KnowledgeContext may for example include:

@@ -87,4 +93,5 @@

-retentionPolicy: This specifies the retention policy of the constituent Knowledge instances. It is defined in terms of hours. The Knowledge older than the defined hours will be deleted. It may indicate the time for which the MnS consumer is recommended to consider the knowledge as useful beyond which time the knowledge can be deleted.

-8.X.3Key Issue #7: Knowledge Lifecycle Management

-8.X.3.1Potential Solution 1: Knowledge and data relationship

+Editor's note: This list of context content can be extended or revised as more key issues are clarified..

+8.X.cPotential Solutions for Key Issue #7: Knowledge Lifecycle Management

+8.X.c.1Potential Solution 1: Knowledge and data relationship

Key issue #7 for knowledge management is Knowledge Lifecycle Management. This key issue, specified in clause 7.2.7 of the present TR, includes two aspects: 1) the lifecycle management of knowledge; and 2) how the lifecycle management of knowledge can be decoupled from the data lifecycle.

@@ -92,5 +99,7 @@

Knowledge can be generated using data. The relationship between data and knowledge can be specified using a mechanism that describes the context related to collected data the knowledge was built on. Specifically, this mechanism consists in providing a snapshot of the configuration of the data collection job(s) that created the data used for generating the knowledge.

-A baseline snapshot includes the following information related to each data collection job.

+Note: Specific knowledge may be created from processing (e.g. aggregating) a large set of data instances/sets. Two knowledge objects associated with the same data instances may not be equivalent

+A snapshot may for example include the following information related to each data collection job.

Data type(s) collected with the job. A data type can be expressed by either its name and/or category.

Collection scope, specifying information related to the geographic area(s) where and time window(s) when the data was collected.

+Editor's note: Whether these or other aspects of data should be the linked information if FFS depending on the stage 2 model for data.

It is worth noting that a data collection job allows collecting data from 3GPP sources and/or non-3GPP sources (e.g., databases).

S5-263547d3 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d4 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d3 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d4 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -50,4 +50,4 @@

The following figure depicts the overall set of entities that could be involved in Knowledge collection, generation and transfer:

-Figure 8.1-1: Knowledge Management deployment options: including where i) 3GPP system provides data and Knowledge production happens in management systems.

-Knowledge MnS Producer: This entity is responsible to generate Knowledge based on the management data received from the management data generator. This entity is also responsible to maintain the Knowledge Base containing the historical knowledge. When the knowledge producer is part of the 3GPP systems, the management data is augmented by the Knowledge and becomes the part of management data only. E.g., in case of PM, the template defined in 28.552 is extended to include Knowledge as part of performance data.

+Figure 8.1-1: Knowledge Management deployment options: including where 3GPP system provides data and Knowledge production happens in management systems.

+Knowledge MnS Producer: This entity is responsible to generate Knowledge based on the management data received from the management data generator. This entity is also responsible to maintain the Knowledge Base containing the historical knowledge.

Knowledge MnS Consumer: This entity is responsible for taking network management decision based on the available knowledge. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

@@ -59,5 +59,5 @@

Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope.

-Editor's Note: Whether Knowledge can be produced and provided by network nodes and network functions is FFS

+Editor's Note: Whether Knowledge can be produced and provided by network element is FFS

2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

-Introduce a KnowledgeJob to represent the process including the MnS consumer's configuration to the MnS producer for producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

+Introduce an optionl KnowledgeJob to represent the process including the MnS consumer's configuration to the MnS producer for producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

3)

@@ -78,3 +78,3 @@

Editor's Note: The nature of the object Knowledge is FFS

-2)Knowledge indicates a graph of relationships between entities. The content to be included as entities or relationships is derived form an ontology.

+2)Knowledge indicates a graph of relationships between entities. The content to be included as entities or relationships is derived from an ontology.

Introduce an ontology in SA5 that contains all vocabulary that can be used aspart of entities or relationships in Knowledge.

@@ -82,3 +82,3 @@

-content from the 3GPP management system NRM

--PMData: which indicates the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

+-Management data, e.g., PMData,.

Example relationship in the ontology may include "isCausedBy", "isCauseOf", "leadTo", "canResultIn", etc.

@@ -93,2 +93,3 @@

-retentionPolicy: This specifies the retention policy of the constituent Knowledge instances. It is defined in terms of hours. The Knowledge older than the defined hours will be deleted. It may indicate the time for which the MnS consumer is recommended to consider the knowledge as useful beyond which time the knowledge can be deleted.

+- kUsedPMData: This defines the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

Editor's note: This list of context content can be extended or revised as more key issues are clarified..

S5-263547d4 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d5 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d4 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d5 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -53,3 +53,3 @@

Knowledge MnS Consumer: This entity is responsible for taking network management decision based on the available knowledge. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

-When the knowledge producer is part of the 3GPP systems, Knowledge is managed as part of Management Data, otherwise when the knowledge producer is part of the 3GPP management systems, Knowledge is managed as part of NRM.

+Knowledge can be managed as part of Management Data or as part of NRM.

8.X.b.2Potential Solution #1: Architectural aspects of Knowledge Management

@@ -61,3 +61,3 @@

2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

-Introduce an optionl KnowledgeJob to represent the process including the MnS consumer's configuration to the MnS producer for producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

+Introduce an optional KnowledgeJob to represent the process including the MnS consumer's configuration to the MnS producer for producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

3)

S5-263547d5 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d6 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d5 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d6 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -51,4 +51,4 @@

Figure 8.1-1: Knowledge Management deployment options: including where 3GPP system provides data and Knowledge production happens in management systems.

-Knowledge MnS Producer: This entity is responsible to generate Knowledge based on the management data received from the management data generator. This entity is also responsible to maintain the Knowledge Base containing the historical knowledge.

-Knowledge MnS Consumer: This entity is responsible for taking network management decision based on the available knowledge. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

+Knowledge MnS Producer: This entity is responsible to generate and provide Knowledge. This entity may maintain a Knowledge Base containing the knowledge.

+Knowledge MnS Consumer: This entity that consumes capabilities offered by knowledge MnS. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

Knowledge can be managed as part of Management Data or as part of NRM.

@@ -58,6 +58,6 @@

Introduce an object representing the capability to provide knowledge-related services, say called KnowledgeFunction.

-Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope.

+Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope or providing input to extend available knowledge.

Editor's Note: Whether Knowledge can be produced and provided by network element is FFS

2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

-Introduce an optional KnowledgeJob to represent the process including the MnS consumer's configuration to the MnS producer for producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

+Introduce an optional KnowledgeJob to represent the MnS consumer's request for and MnS producer process of producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

3)

@@ -75,3 +75,3 @@

The following is proposed to be introduced in SA5 to support Knowledge representation.

-1)Knowledge can be delivered to the MnS consumer, e.g., via a report.

+1)Knowledge can be delivered to the MnS consumer.

Introduce an object representing the knowledge, say called Knowledge. The Knowledge is associated with a KnowledgeFunction, a KnowledgeJob .

@@ -87,2 +87,3 @@

Knowledge should include the relationships among the entities. Introduce an attribute on Knowledge to represent the relations, e.g., to be called KnowledgeRelation.

+Editor's note: The definition and detailed modelling of KnowledgeNode and KnowledgeRelation can be revisited as part of the ontology solution

3)Knowledge can be associated with specific context that helps in its interpretability. Introduce an attribute for context, to be called KnowledgeContext. The KnowledgeContext may for example include:

S5-263547d6 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d7 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d6 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d7 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -53,3 +53,2 @@

Knowledge MnS Consumer: This entity that consumes capabilities offered by knowledge MnS. The role of this entity can be played by existing management functions e.g., CCL. A CCL can utilize available Knowledge to take configuration decision as per the execution step of CCL stages.

-Knowledge can be managed as part of Management Data or as part of NRM.

8.X.b.2Potential Solution #1: Architectural aspects of Knowledge Management

@@ -59,7 +58,5 @@

Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope or providing input to extend available knowledge.

-Editor's Note: Whether Knowledge can be produced and provided by network element is FFS

-2)Knowledge may be requested by an MnS consumer from the MnS producer either to be provided for a single shot or to be continuously provided for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

-Introduce an optional KnowledgeJob to represent the MnS consumer's request for and MnS producer process of producing knowledge. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

-3)

-The knowledge production object (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify the scope for which knowledge can be provided

+2)A Knowledge MnS consumer can query the MnS producer to deliver knowledge or can contribute knowledge to the MnS producer. The delivery or contribution can either to be provided for a single shot or be continuous for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

+Introduce an optional KnowledgeJob to represent the MnS consumer's request to the MnS producer. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

+3)The knowledge production object (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify the scope for which knowledge can be provided

Introduce a set of configurable attributes on the KnowledgeFunction. Example configurable attributes include:

@@ -71,3 +68,2 @@

Editor's note: the list of attributes of the KnowledgeJob can be extended

-Editor's note: Whether the KnowledgeJob should allow the MnS consumer to provide knowledge to the MnS producer is FFS.

8.X.aPotential Solutions for Key Issue #1: Knowledge Representation and Management

S5-263547d7 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d8 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d7 (was S5-262930) Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d8 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -1,4 +1,4 @@

-3GPP TSG-SA5 Meeting #168S5-262930

+3GPP TSG-SA5 Meeting #168S5-263547

Prague, Czech Republic, 24 – 28 August 2026

-Source:Nokia (Moderator)

+Source:Nokia (Moderator), Samsung,

Title:Pseudo-CR on Solutions Knowledge Architecture & Modelling

@@ -58,3 +58,3 @@

Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope or providing input to extend available knowledge.

-2)A Knowledge MnS consumer can query the MnS producer to deliver knowledge or can contribute knowledge to the MnS producer. The delivery or contribution can either to be provided for a single shot or be continuous for as long certain conditions are true. The MnS consumer may want to track the process of providing that knowledge.

+2)A Knowledge MnS consumer can request the MnS producer to provide knowledge or can contribute knowledge to the MnS producer. The delivery or contribution can either to be provided for a single shot or be continuous for as long certain conditions are true. The MnS consumer may want to track the progress of the request.

Introduce an optional KnowledgeJob to represent the MnS consumer's request to the MnS producer. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

S5-263547d8 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy.docx

--- S5-263547d8 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy.docx

@@ -96,2 +96,3 @@

The present solution addresses the second aspect, by focusing on relationship between knowledge and data.

+-----------------------E// Version-----------------------------

Knowledge can be generated using data. The relationship between data and knowledge can be specified using a mechanism that describes the context related to collected data the knowledge was built on. Specifically, this mechanism consists in providing a snapshot of the configuration of the data collection job(s) that created the data used for generating the knowledge.

@@ -103,2 +104,4 @@

It is worth noting that a data collection job allows collecting data from 3GPP sources and/or non-3GPP sources (e.g., databases).

+--------------------------SS Version--------------------------------------

+Knowledge can be generated using data. The relationship between data and knowledge can be specified using a knowledge graph and/or data used for generating the knowledge

* * * End of Changes * * * *

S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy.docx -> S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy-20260828_online.docx

--- S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy.docx

+++ S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy-20260828_online.docx

@@ -60,2 +60,4 @@

Introduce an optional KnowledgeJob to represent the MnS consumer's request to the MnS producer. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

+2)A Knowledge MnS consumer can request to consume knowledge from or to contribute knowledge to the MnS producer. The delivery or contribution can either to be provided for a single shot or be continuous for as long certain conditions are true. The MnS consumer may want to track the progress of the request.

+2)A Knowledge MnS consumer can request to consume knowledge from MnS producer. A Knowledge MnS consumer can request to contribute to the knowledge produced by the MnS producer.

3)The knowledge production object (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify the scope for which knowledge can be provided

S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy-20260828_online.docx -> S5-263547d10 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d9 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling - Copy-20260828_online.docx

+++ S5-263547d10 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -58,6 +58,4 @@

Editor's note: The exact list of management capabilities to be provided to the MnS consumers is FFS. Examples may include discovering available knowledge and requesting for knowledge fitted to a given scope or providing input to extend available knowledge.

-2)A Knowledge MnS consumer can request the MnS producer to provide knowledge or can contribute knowledge to the MnS producer. The delivery or contribution can either to be provided for a single shot or be continuous for as long certain conditions are true. The MnS consumer may want to track the progress of the request.

-Introduce an optional KnowledgeJob to represent the MnS consumer's request to the MnS producer. The KnowledgeJob allows for single shot or continuous delivery of knowledge.

-2)A Knowledge MnS consumer can request to consume knowledge from or to contribute knowledge to the MnS producer. The delivery or contribution can either to be provided for a single shot or be continuous for as long certain conditions are true. The MnS consumer may want to track the progress of the request.

-2)A Knowledge MnS consumer can request to consume knowledge from MnS producer. A Knowledge MnS consumer can request to contribute to the knowledge produced by the MnS producer.

+2)A Knowledge MnS producer should support the retrieval of knowledge and allow contributions towards knowledge.

+Introduce an optional KnowledgeJob to represent the consumer's request to retrieve or contribute. Asynchronous / synchronous jobs can be supported

3)The knowledge production object (i.e., KnowledgeFunction) can be configured, e.g. by the operator to specify the scope for which knowledge can be provided

@@ -78,10 +76,4 @@

Introduce an ontology in SA5 that contains all vocabulary that can be used aspart of entities or relationships in Knowledge.

-Example content for the entities in the ontology could be:

--content from the 3GPP management system NRM

--Management data, e.g., PMData,.

Example relationship in the ontology may include "isCausedBy", "isCauseOf", "leadTo", "canResultIn", etc.

Editor's note: The full list of items to be included in the vocabulary in the ontology are FFS.

-Knowledge should include the entities that are being related as vertices of the graph. Introduce an attribute on Knowledge to represent the vertex, e.g., to be called KnowledgeNode. The KnowledgeNode may for example be instance of data, event, etc.

-Knowledge should include the relationships among the entities. Introduce an attribute on Knowledge to represent the relations, e.g., to be called KnowledgeRelation.

-Editor's note: The definition and detailed modelling of KnowledgeNode and KnowledgeRelation can be revisited as part of the ontology solution

3)Knowledge can be associated with specific context that helps in its interpretability. Introduce an attribute for context, to be called KnowledgeContext. The KnowledgeContext may for example include:

@@ -92,18 +84,3 @@

-retentionPolicy: This specifies the retention policy of the constituent Knowledge instances. It is defined in terms of hours. The Knowledge older than the defined hours will be deleted. It may indicate the time for which the MnS consumer is recommended to consider the knowledge as useful beyond which time the knowledge can be deleted.

-- kUsedPMData: This defines the PM data that is used to derive this information. This will be the PMs and KPIs name as defined in 3GPP TS 28.552 and 28.554 respectively.

Editor's note: This list of context content can be extended or revised as more key issues are clarified..

-8.X.cPotential Solutions for Key Issue #7: Knowledge Lifecycle Management

-8.X.c.1Potential Solution 1: Knowledge and data relationship

-Key issue #7 for knowledge management is Knowledge Lifecycle Management. This key issue, specified in clause 7.2.7 of the present TR, includes two aspects: 1) the lifecycle management of knowledge; and 2) how the lifecycle management of knowledge can be decoupled from the data lifecycle.

-The present solution addresses the second aspect, by focusing on relationship between knowledge and data.

------------------------E// Version-----------------------------

-Knowledge can be generated using data. The relationship between data and knowledge can be specified using a mechanism that describes the context related to collected data the knowledge was built on. Specifically, this mechanism consists in providing a snapshot of the configuration of the data collection job(s) that created the data used for generating the knowledge.

-Note: Specific knowledge may be created from processing (e.g. aggregating) a large set of data instances/sets. Two knowledge objects associated with the same data instances may not be equivalent

-A snapshot may for example include the following information related to each data collection job.

-Data type(s) collected with the job. A data type can be expressed by either its name and/or category.

-Collection scope, specifying information related to the geographic area(s) where and time window(s) when the data was collected.

-Editor's note: Whether these or other aspects of data should be the linked information if FFS depending on the stage 2 model for data.

-It is worth noting that a data collection job allows collecting data from 3GPP sources and/or non-3GPP sources (e.g., databases).

---------------------------SS Version---------------------------------------

-Knowledge can be generated using data. The relationship between data and knowledge can be specified using a knowledge graph and/or data used for generating the knowledge

* * * End of Changes * * * *

S5-263547d10 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx -> 263547_final_S5-263547 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

--- S5-263547d10 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

+++ 263547_final_S5-263547 Rel-20 pCR TR32.801 KSM3 - Solutions on Knowledge Architecture & Modelling.docx

@@ -81,3 +81,3 @@

-Time: This defines the time e.g., at which the knowledge was generated or would apply.

--Purpose: This defines the network problem categories where this knowledge can be applied. Example purposes may include: COVERAGE, MOBILITY, SOFTWARE_UPGRADE, RESOURCE_DEPLITION.

+-Purpose: This defines the network problem categories where this knowledge can be applied. Example purposes may include: COVERAGE, MOBILITY, SOFTWARE_UPGRADE, RESOURCE_DEPLETION.

Editor's Note: Whether specific values for Purpose should specified e.g., as ENUM is FFS

S5-263620 - Cloud管理讨论稿

S5-263048 Discussion Paper on Cloud Management in 6G.docx -> S5-263048r3 Discussion Paper on Cloud Management in 6G.docx

--- S5-263048 Discussion Paper on Cloud Management in 6G.docx

+++ S5-263048r3 Discussion Paper on Cloud Management in 6G.docx

@@ -1,2 +1,2 @@

-3GPP TSG-SA5 Meeting #168S5-263048

+3GPP TSG-SA5 Meeting #168S5-2630482r3

Prague, Czech Republic, 24-28 August 2026

@@ -26,3 +26,3 @@

NOTE: The analysis in SA2 TR 23.801-01 has not been concluded and the information above is subject to change.

-Challenge#1: The need for separation between 3GPP Management in SA5 and Cloud Management

+Challenge: The need for separation between 3GPP Management in SA5 and Cloud Management

In principle, cloud management systems provide generic cloud capabilities including workload lifecycle management, scheduling, resource management and resource allocation, scaling, networking, storage management, observability, security, and infrastructure automation. These capabilities are widely implemented by cloud ecosystems and continue to evolve independently of mobile network standards.

@@ -53,6 +53,2 @@

Currently, up to 5GA, the NF Deployment model for cloudified NFs follows Option#2.

-Challenge#2: Enhancements to the object modelling considered for cloudified 6G NFs

-The ManagedFunction IoC defined in TS 28.622 specifies the attribute vnfParametersList used when the ManagedFunction instance is realized by one or more VNF instances.

-The vnfParametersList currently specifies one mandatory attribute, vnfInstanceId, which is a VNF instance identifier according to clause 9.4.2 of ETSI GS NFV-IFA 008, and the following optional attributes: vnfdId, which identifies the VNFD on which the VNF instance is based; flavourId, which identifies the VNF deployment flavour applied to the VNF instance; and autoScalable, which indicates whether auto-scaling of the VNF instance is enabled or disabled.

-To better support modern 6G use cases, it is expected that similar or additional attributes might be specified for NF Deployments.

4Detailed proposals

@@ -61,6 +57,4 @@

Proposal#1.1: As in 5GA, the 3GPP management system in 6G needs to interact with external orchestration and management systems via the Deployment management reference point to support lifecycle management (LCM) of NF Deployment instances.

-Proposal#1.2: In 6G, the 3GPP management system needs to interact with external orchestration and management systems via the NF Deployment management reference point (or a new management reference point) to support OAM of NF Deployment instances.

+Proposal#1.2: In 6G, the 3GPP management system might need to interact with external orchestration and management systems to support observability and provisioning of NF Deployment instances.

NOTE: Depending on the analysis, requirements and solutions endorsed by the 6G study for 3GPP SA5-defined applications other than NFs that are deployed in cloudified environments, a similar approach can be considered.

-Challenge#2: Modelling of NF Deployments in 6G:

-Proposal#2.1: For NF Deployments, use as a baseline the modelling approach followed for VNFs defined in TS 28.622 (with a model similar to the vnfParametersList). Introduce additional attributes if needed based on the analysis, requirements, and solutions endorsed by the 6G study.

The group is asked to endorse the proposals above.

S5-263048r3 Discussion Paper on Cloud Management in 6G.docx -> S5-263620d1 Discussion Paper on Cloud Management in 6G.docx

--- S5-263048r3 Discussion Paper on Cloud Management in 6G.docx

+++ S5-263620d1 Discussion Paper on Cloud Management in 6G.docx

@@ -1,2 +1,2 @@

-3GPP TSG-SA5 Meeting #168S5-2630482r3

+3GPP TSG-SA5 Meeting #168S5-263620d1

Prague, Czech Republic, 24-28 August 2026

@@ -55,6 +55,5 @@

Challenge#1: Interactions with external cloud management systems for NFs deployed in cloudified environments (NF Deployments) in 6G:

-The 3GPP management system should focus on telecommunications-specific management (services, network functions, management applications, policies, performance, and fault management), while cloud management systems remain responsible for cloud infrastructure, workload orchestration, scheduling, scaling, and resource allocation. This separation avoids overlapping responsibilities and duplicate standardization efforts and provides a clear scope of activities for 3GPP SA5.

-Proposal#1.1: As in 5GA, the 3GPP management system in 6G needs to interact with external orchestration and management systems via the Deployment management reference point to support lifecycle management (LCM) of NF Deployment instances.

-Proposal#1.2: In 6G, the 3GPP management system might need to interact with external orchestration and management systems to support observability and provisioning of NF Deployment instances.

-NOTE: Depending on the analysis, requirements and solutions endorsed by the 6G study for 3GPP SA5-defined applications other than NFs that are deployed in cloudified environments, a similar approach can be considered.

+The 3GPP management system should focus on telecommunications-specific management (services, network functions, management applications, policies, performance, and fault management), while cloud management systems remain responsible for cloud infrastructure, workload orchestration, scheduling, and resource allocation. This separation avoids overlapping responsibilities and duplicate standardization efforts and provides a clear scope of activities for 3GPP SA5.

+Proposal#1.1: As in 5GA according to TS28.531 and TS28.533, the 3GPP management system in 6G needs to interact with external orchestration and management systems via the Deployment management reference point to support lifecycle management (LCM) of NF Deployment instances.

+Proposal#1.2: In 6G, the 3GPP management system might need to interact with external orchestration and management systems to support observability and configuration of NF Deployment instances.

The group is asked to endorse the proposals above.

S5-263620d1 Discussion Paper on Cloud Management in 6G.docx -> S5-263620d2 Discussion Paper on Cloud Management in 6G.docx

--- S5-263620d1 Discussion Paper on Cloud Management in 6G.docx

+++ S5-263620d2 Discussion Paper on Cloud Management in 6G.docx

@@ -1,2 +1,2 @@

-3GPP TSG-SA5 Meeting #168S5-263620d1

+3GPP TSG-SA5 Meeting #168S5-263620d2

Prague, Czech Republic, 24-28 August 2026

@@ -58,2 +58,3 @@

Proposal#1.2: In 6G, the 3GPP management system might need to interact with external orchestration and management systems to support observability and configuration of NF Deployment instances.

+NOTE: For 3GPP SA5-defined applications other than NFs that are deployed in cloudified environments, the ongoing 6G study is ongoing. Depending on the analysis, requirements and solutions that are endorsed, a similar approach can be considered.

The group is asked to endorse the proposals above.

S5-263620d2 Discussion Paper on Cloud Management in 6G.docx -> S5-263620d3 Discussion Paper on Cloud Management in 6G.docx

--- S5-263620d2 Discussion Paper on Cloud Management in 6G.docx

+++ S5-263620d3 Discussion Paper on Cloud Management in 6G.docx

@@ -1,2 +1,2 @@

-3GPP TSG-SA5 Meeting #168S5-263620d2

+3GPP TSG-SA5 Meeting #168S5-263620d3

Prague, Czech Republic, 24-28 August 2026

@@ -58,3 +58,3 @@

Proposal#1.2: In 6G, the 3GPP management system might need to interact with external orchestration and management systems to support observability and configuration of NF Deployment instances.

-NOTE: For 3GPP SA5-defined applications other than NFs that are deployed in cloudified environments, the ongoing 6G study is ongoing. Depending on the analysis, requirements and solutions that are endorsed, a similar approach can be considered.

+NOTE: For 3GPP SA5-defined applications other than NFs that are deployed in cloudified environments, the 6G study is ongoing. Depending on the analysis, requirements and solutions that are endorsed, a similar approach can be considered.

The group is asked to endorse the proposals above.

S5-263620d3 Discussion Paper on Cloud Management in 6G.docx -> 263620_final_S5-263620 Discussion Paper on Cloud Management in 6G.docx

--- S5-263620d3 Discussion Paper on Cloud Management in 6G.docx

+++ 263620_final_S5-263620 Discussion Paper on Cloud Management in 6G.docx

@@ -1,2 +1,2 @@

-3GPP TSG-SA5 Meeting #168S5-263620d3

+3GPP TSG-SA5 Meeting #168S5-263620

Prague, Czech Republic, 24-28 August 2026

S5-263621 - 声明式管理术语

S5-263229 Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx -> S5-263621d1 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

--- S5-263229 Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

+++ S5-263621d1 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263229

+3GPP TSG SA5 Meeting #168S5-263621

Prague, Czech Republic, 24-28 August 2026

@@ -16,18 +16,19 @@

The deployment and lifecycle management (LCM) of NF Deployment require innovative approaches that ensure agility, consistency, automation, and full traceability. Traditional management methods, which rely heavily on manual interventions and imperative workflows, are insufficient to meet the demands of 6G networks characterized by massive scale, dynamic service requirements, and continuous evolution.

-The Declarative approach provides a, version-controlled, and automated framework for managing the entire lifecycle of NF Deployment. In this paradigm, the desired state of network functions—including configurations, deployment parameters, policies, and dependencies—is defined declaratively in a version-controlled source-of-truth system.

-GitOps is an example realization of Declarative approach, where Git repositories are commonly used as the source-of-truth system. However, the Declarative approach does not require the use of Git, Git repositories, or Git-specific protocols.

+The declarative approach provides a, version-controlled, and automated framework for managing the entire lifecycle of NF Deployment. In this paradigm, the desired state of network functions—including configurations, deployment parameters, policies, and dependencies—is defined declaratively in a version-controlled source-of-truth system.

+A declarative approach defines the desired end state, rather than specifying the steps needed to achieve it.

+GitOps is an example realization of declarative approach, where Git repositories are commonly used as the source-of-truth system. However, the declarative approach does not require the use of Git, Git repositories, or Git-specific protocols.

Editor’s Note: NF and NF Deployment is not the same. What is the declared state of NF vs. declared state of NF Deployment? Need to elaborate.

-In principle, the Declarative paradigm can be used to support the NF Deployment related management operations. Interactions with a Declarative system can be seen as a potential solution for the generic deployment reference point described in TR 28.869 to support NF Deployments, through interactions between the 3GPP management system and external Declarative based mechanisms.

+In principle, the declarative paradigm can be used to support the NF Deployment related management operations. Interactions with a declarative system can be seen as a potential solution for the generic deployment reference point described in TR 28.869 to support NF Deployments, through interactions between the 3GPP management system and external declarative based mechanisms.

Network Functions Deployment:

For NFs deployed in the cloud, deployment descriptors, package manifests, and configuration files are stored in a version-controlled source-of-truth system, serving as the single source of truth.

-A Declarative approach can be used to automate the deployment pipeline, the provisioning, configuration, and instantiation of NF Deployment across distributed cloud environments (centralized cloud, edge cloud, distributed edge).

-A Declarative approach can be used to support continuous Integration/Continuous Deployment (CI/CD) pipelines to enable rapid, reliable, and repeatable deployments.

-Declarative based deployments can be further enhanced by integrating with other compatible toolboxes, like container registries, package registries, testing utilities, and CI support to facilitate the deployment process.

+A declarative approach can be used to automate the deployment pipeline, the provisioning, configuration, and instantiation of NF Deployment across distributed cloud environments (centralized cloud, edge cloud, distributed edge).

+A declarative approach can be used to support continuous Integration/Continuous Deployment (CI/CD) pipelines to enable rapid, reliable, and repeatable deployments.

+declarative based deployments can be further enhanced by integrating with other compatible toolboxes, like container registries, package registries, testing utilities, and CI support to facilitate the deployment process.

Lifecycle Management (LCM) of NF Deployment:

-Declarative procedures can be used to manage the complete lifecycle of NF Deployment, including instantiation, scaling, upgrading, and termination.

-In Declarative approaches, lifecycle operations are triggered declaratively through changes to desired-state information, ensuring full traceability and auditability.

-In Declarative approaches, automatically restore to previous known configuration issupported, allowing rapid recovery from failed updates or misconfigurations.

-Declarative approaches can be used to coordinate with underlying cloud infrastructure and orchestration layers (e.g., NFVO, VNFM) to execute lifecycle operations seamlessly.

+declarative procedures can be used to manage the complete lifecycle of NF Deployment, including instantiation, scaling, upgrading, and termination.

+In declarative approaches, lifecycle operations are triggered declaratively through changes to desired-state information, ensuring full traceability and auditability.

+In declarative approaches, automatically restore to previous known configuration issupported, allowing rapid recovery from failed updates or misconfigurations.

+declarative approaches can be used to coordinate with underlying cloud infrastructure and orchestration layers (e.g., NFVO, VNFM) to execute lifecycle operations seamlessly.

Real-Time Monitoring, Validation, and Self-Healing:

-Automated validation checks ensure that the actual state matches the desired state defined in the single source of truth for the Declarative system.

+Automated validation checks ensure that the actual state matches the desired state defined in the single source of truth for the declarative system.

Policy-Driven Management and Governance:

@@ -36,16 +37,16 @@

Every change to NF configurations, deployments, or policies is recorded in the Declarative system, providing a complete audit trail.

-Declarative procedures can be used to generate detailed reports on deployment history, lifecycle events, supporting post-deployment analysis and continuous improvement.

+declarative procedures can be used to generate detailed reports on deployment history, lifecycle events, supporting post-deployment analysis and continuous improvement.

Security and Artifact Management:

Artifact management is centralized, with support for container images, Helm charts, and other cloud-native packages.

-Declarative procedures can support artifact verification, and supply chain security to ensure only validated and approved artifacts are deployed.

+declarative procedures can support artifact verification, and supply chain security to ensure only validated and approved artifacts are deployed.

Continuous Testing:

-Declarative procedures can support continuous testing of deployments, integrating testing utilities and frameworks.

+declarative procedures can support continuous testing of deployments, integrating testing utilities and frameworks.

Test results are stored and exposed through APIs and databases, enabling continuous validation and quality assurance.

Testing is linked to the continuous deployment process, ensuring that every deployment is accompanied by appropriate validation.

-Editor’ s Note: Declarative procedures can be also used to support internal 3GPP management system procedures but there is no related analysis. This is FFS.

-Editor’ s Note: the description above is about Declarative approach as an alternative solution to the generic reference point in TR28.869 to support NF Deployments but there is no description about the relationship with the 3GPP management system rather general concepts about using Declarative approach for cloudified functions management. It is FFS to describe the relationship with SBMA and to determine the proper clause in the TR where these general Declarative approach concepts will be captured.

+Editor’ s Note: declarative procedures can be also used to support internal 3GPP management system procedures but there is no related analysis. This is FFS.

+Editor’ s Note: the description above is about declarative approach as an alternative solution to the generic reference point in TR28.869 to support NF Deployments but there is no description about the relationship with the 3GPP management system rather general concepts about using declarative approach for cloudified functions management. It is FFS to describe the relationship with SBMA and to determine the proper clause in the TR where these general declarative approach concepts will be captured.

6.1.7.3.2Potential Requirements

-REQ-6G-DECL-01: The 6G management system should support interaction with Declarative based external management and orchestration systems for the management of NF Deployments (e.g., instantiation, upgrading, termination, configuration).

-Note: NF deployment management is through the generic reference point defined in TS 28.533 [5]. A Declarative based solution, such as a GitOps-based solution, is one possible implementation

-REQ-6G-DECL-02: The 6G management system should support interactions with Declarative based external management and orchestration systems for the traceability and auditability of NF deployments.

+REQ-6G-DECL-01: The 6G management system should support interaction with declarative based external management and orchestration systems for the management of NF Deployments (e.g., instantiation, upgrading, termination, configuration).

+Note: NF deployment management is through the generic reference point defined in TS 28.533 [5]. A declarative based solution, such as a GitOps-based solution, is one possible implementation

+REQ-6G-DECL-02: The 6G management system should support interactions with declarative based external management and orchestration systems for the traceability and auditability of NF deployments.

* * * End of Changes * * * *

S5-263621d1 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx -> S5-263621d2 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

--- S5-263621d1 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

+++ S5-263621d2 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

@@ -36,3 +36,3 @@

Traceability, Auditability, and Continuous Improvement:

-Every change to NF configurations, deployments, or policies is recorded in the Declarative system, providing a complete audit trail.

+Every change to NF configurations, deployments, or policies is recorded in the declarative system, providing a complete audit trail.

declarative procedures can be used to generate detailed reports on deployment history, lifecycle events, supporting post-deployment analysis and continuous improvement.

S5-263621d2 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx -> 263621_final_S5-263621d2 (was 3229) Rel-20 pCR 32.801-01 Technology Neutral terminology for GitOps based use-case.docx

NO TEXT CHANGE

S5-263622 - NF部署LCM与NRM

S5-263119 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx -> S5-263622d1 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

--- S5-263119 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

+++ S5-263622d1 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263119

+3GPP TSG SA5 Meeting #168S5-263622

Prague, Czech Republic, 24-28 August 2026

@@ -49,7 +49,7 @@

The scope of the Rel-19 study conducted on LCM of NF Deployments did not include the impact of NF Deployments on higher layer use cases like Sub-Network, Network Slice Subnets.

-As part of the normative work in Rel-20, the usage of Deployment management reference point has been specified in TS 28.533 [5] to support lifecycle management (LCM) of NF Deployment instances. Updated use cases, requirements and procedure for LCM of NF instances realized by cloud-native deployments are now specified in TS 28.531 [4]. In addition, the updated procedures for Network Slice Subnet have been specified. The lifecycle impacts of 3GPP sub-networks for NF deployments, not addressed in 5G/5GA, needs to be studied as part of 6G.

-Additionally, the Network Resource Model needs to be re-examined with regards to NF Deployments. As per TS 28.662, clause 4.4.3, ManagedElement represents a telecommunication equipment which may contain hardware and software components. In the case of cloud-based instances, it will present the software component only and such an instance is known as NF Deployment. A ManagedElement may contain several managed functions. However, the ManagedElement IOC is not associated with a cloud software instance such as VNF. Currently, the vnfParametersList attribute is a part of the ManagedFunction IOC which causes confusion and makes it difficult to associate a single VNF with multiple ManagedFunction instances. The relationship between ManagedFunction/ManagedElement and the existing VNFParametersList and the newly defined NF Deployments needs to be further examined in 6G OAM studies.

+As part of the normative work in Rel-20, the usage of Deployment management reference point has been specified in TS 28.533 [5] to support lifecycle management (LCM) of NF Deployment instances. Updated use cases, requirements and procedure for LCM of NF instances realized by NF Deployments are now specified in TS 28.531 [4]. In addition, the updated procedures for Network Slice Subnet have been specified. The lifecycle impacts of 3GPP sub-networks for NF deployments, not addressed in 5G/5GA, needs to be studied as part of 6G.

+Additionally, the Network Resource Model needs to be re-examined with regards to NF Deployments. As per TS 28.662, clause 4.4.3, ManagedElement represents a telecommunication equipment which may contain hardware and software components. In the case of cloud-based instances, the ME will represent the virtualized software component only and such an instance is also known as NF Deployment. A ManagedElement may contain several managed functions. However, the ManagedElement IOC is not associated with a cloud software instance such as VNF. Currently, the vnfParametersList attribute is a part of the ManagedFunction IOC which causes confusion and makes it difficult to associate a single VNF with multiple ManagedFunction instances. The relationship between ManagedFunction/ManagedElement and the existing VNFParametersList and the newly defined NF Deployments needs to be further examined in 6G OAM studies.

6.1.7.2.2Potential Requirements

REQ-CMO-1: 3GPP management system should support LCM of Sub-Network which include 3GPP NF deployed as NF Deployments.

-REQ-CMO-2: 3GPP management system should support an association between ManagedElement and NF Deployments.

+REQ-CMO-A: 3GPP management system should support modelling improvements for NF Deployments.

* * * Next Change * * * *

@@ -59,175 +59,15 @@

7.5.XKey Issue #X: NRM Improvements for NF Deployments

-6G management system should support an improved NRM which establishes a relationship between ManagedElement IOC and NF Deployments hosted on it. These NF Deployments may be associated with one or more ManagedFunction instances across both 5G and 6G environments, addressing the gaps not covered by 5G specifications. This key issue is related to the requirements specified in REQ-CMO-2.

+This key issue is related to the requirements specified in REQ-CMO-A and focusses on the following aspects

+1. How NRM can be improved to support NF Deployments?

+2. Whether and how an association is described between ManagedElement IOC and a potential NF Deployment-related IOC?

+3. Whether and how the association is described between ManagedFunction IOC and a potential NF Deployment related IOC?

+4. Whether the existing vnfParametersList attributes needs to be deprecated?

* * * Next Change * * * *

8.YPotential Solution #Y: Updates to LCM of 3GPP sub-network using NF Deployments

-This potential solution addresses the Key Issue defined in clause 7.5.1. Currently, TS 28.531 [4] has defined use cases for creation and configuration of 3GPP sub-network. The relevant usecases for sub-network (namely 5.1.19 and 5.1.20) can be potentially updated as follows.

+This potential solution addresses the Key Issue defined in clause 7.5.1.

+Updates for Creation of a 3GPP Sub-network:

+The existing use case in section 5.1.19 of TS 28.531 [4] describes the end-to-end provisioning of a new 3GPP sub-network through a NF provisioning management service. Since the usage of NF Deployments is now specified in 5GA, this use case needs to limit the instantiation of NSs to ETSI NFV-MANO specific cases. Additionally, an alternative procedure would need to be defined based on other industry solutions. If necessary, additional NF Deployments for sub-networks may be created via the Deployment Management reference point.

+Updates for Configuration of a 3GPP Sub-network:

+The existing use case in section 5.1.20 of TS 28.531 [4] describes the modification of 3GPP sub-network through a NF provisioning management service. Since the usage of NF Deployments is now specified in 5GA, this use case needs to limit the update of NSs to ETSI NFV-MANO specific cases. Additionally, an alternative procedure would need to be defined based on other industry solutions. If necessary, additional NF Deployments for sub-networks may be created or existing ones modified via the Deployment Management reference point.

Editor’ s Note: It is FFS if detailed procedures are to be described for LCM of 3GPP sub-networks while using NF Deployments

-5.1.19Creation of a 3GPP sub-network

-Use case stage

-Evolution/Specification

-<<Uses>>Related use

-Goal

-To enable the authorized consumer to request creation of a 3GPP sub-network.

-Actors and Roles

-An authorized consumer of the sub-network provisioning management service.

-Telecom resources

-External orchestration and management system,

-If ETSI NFV-MANO system is used,

-VNF package(s) of the virtualized part of 3GPP NF(s);

-NSD(s) of the NS(s);

-Network provisioning service producer;

-NF provisioning service producer.

-Assumptions

-N/A

-Pre-conditions

-The non-virtualized part of the NFs (including completely non-virtualized NFs) constituting the 3GPP sub-network have been deployed;

-If ETSI NFV-MANO system is used,

-The VNF package(s) of the virtualized part of 3GPP NF(s) have been on-boarded to ETSI NFV MANO system;

-The NSD(s) of the NS realizing the 3GPP sub-network have been on-boarded to ETSI NFV MANO system.

-Begins when

-The authorized consumer needs to create a 3GPP sub-network.

-Step 1 (M)

-The authorized consumer requests the sub-network provisioning management service producer to create a 3GPP sub-network. The request needs to indicate the network capacity (e.g., the number of instances of each kind of NFs, and the capacity of each NF instance, for example, number of flows with certain QoS attributes to be supported), network topology information (e.g., the connections between NF instances), and the network QoS requirements (e.g., bandwidth and latency requirements of the interface between two NF instances).

-Step 2 (M)

-The network provisioning management service producer interacts, or requests another network provisioning management service producer to interact, with an external orchestration and management entity.

-In the case of ETSI NFV-MANO, NS(s) is instantiated to realize the sub-network.

-In case of other industry solutions, the network LCM operations are dependent on the capabilities provided by the external orchestration and management system

-Step 3 (M)

-In the case of ETSI NFV MANO, it informs the NF provisioning management service producer about the instantiation of the NSs and the new VNFs.

-Step 4 (M)

-The NF provisioning management configuration service producer creates the MOI(s) of the 3GPP NFs that are realized by the newly instantiated NFDeployment instance(s). There may be MOI(s) that specify the topology of the instantiated ETSI NFV-MANO NSs or its equivalent.

-Step 5 (M)

-The sub-network provisioning management service producer is using the NF provisioning management service to configure the 3GPP NF instance(s) that are constituting the subject 3GPP sub-network.

-NF configuration service

-Ends when

-All the steps identified above are successfully completed.

-Exceptions

-One of the steps identified above fails.

-Post-conditions

-The 3GPP sub-network has been created.

-Traceability

-REQ-PRO_NW-FUN-1, REQ-PRO_NW-FUN-2

-5.1.20Configuration of a 3GPP sub-network

-Use case stage

-Evolution/Specification

-<<Uses>>Related use

-Goal

-To enable the authorized consumer to request configuration of a 3GPP sub-network.

-Actors and Roles

-An authorized consumer of the network provisioning management service.

-Telecom resources

-3GPP network;

-3GPP NFs;

-External orchestration and management system;

-Network provisioning management service producer.

-Assumptions

-N/A

-Pre-conditions

-The 3GPP sub-network has been created;

-The MOI(s) related to the sub-network has been created.

-Begins when

-The authorized consumer needs to configure a 3GPP sub-network.

-Step 1 (M)

-The authorized consumer requests to configure a 3GPP sub-network.

-Step 2 (M)

-The consumer requests the network provisioning management service producer to modify the attribute of the MOI(s) related to the 3GPP sub-network.

-Step 3 (O)

-If the 3GPP network is realized by using a ETSI NFV-MANO NS(s), the network provisioning management service producer requests (directly or indirectly via another) ETSI NFV MANO system to update the NS(s) realizing the 3GPP sub-network.

-In case of other industry solutions, the network LCM operations are dependent on the capabilities provided by the external orchestration and management system

-Step 4 (O)

-In the case of ETSI NFV-MANO, if there are new VNFs instantiated by the NS update, ETSI NFV MANO system informs the NF provisioning management service producer about the instantiation of VNFs.

-Step 5 (O)

-The NF provisioning management service producer creates the MOI(s) of the 3GPP NFs that are realized by the newly instantiated VNF(s).

-Step 6 (M)

-The network provisioning management service producer consumes the NF provisioning management service to configure the impacted 3GPP NF instance(s).

-NF configuration service

-Step 7 (M)

-The network provisioning management service producer configures the 3GPP sub-network, per the MOI attribute modification request received from the consumer.

-Step 8 (M)

-The NF provisioning management service producer modifies the attributes of the MOI(s) of the 3GPP network and informs the consumer that the 3GPP sub-network has been configured successfully.

-Ends when

-All the steps identified above are successfully completed.

-Exceptions

-One of the steps identified above fails.

-Post-conditions

-The 3GPP network has been configured.

-Traceability

-REQ-PRO_NW-FUN-3, REQ-PRO_NW-FUN-4

-* * * Next Change * * * *

-8.ZPotential Solution #Z: Updates to ManagedElement IOC using NF Deployments

-This potential solution addresses the Key Issue defined in clause 7.5.X. The current relationships in NRM between ManagedElement, ManagedFunction and the attribute vnfParametersList is shown in Figure 8.Z-1 below.

-Figure 8.Z-1: Current NRM relationships related to VNF instances

-In order to support an improved NRM for the recently defined NF Deployments, a new object named NFDeploymentParameters can be defined. In addition, ManagedElement IOC and ManagedFunction IOC can be updated with list of NFDeploymentParameters. The updated NRM would look as shown below in Figure 8.Z-2.

-Editor’ s Note: In Figure 8.Z-2, VnfParametersList has been removed to enhance clarity. However, it is FFS if this attribute is to be deprecated or kept as it is for backward compatibility

-Figure 8.Z-2: Proposed NRM relationships related to NF Deployments

-A new attribute named nfDeploymentParametersList can be added to the ManagedElement IOC defined in clause 4.3.3. The updated attribute for ManagedElement IOC are shown below.

-Attribute Name

-S

-isReadable

-isWritable

-isInvariant

-isNotifyable

-vendorName

-M

-T

-F

-F

-T

-userDefinedState

-M

-T

-T

-F

-T

-swVersion

-M

-T

-F

-F

-T

-priorityLabel

-O
-T
-T
-F
-T
-supportedPerfMetricGroups
-O
-T
-F
-F
-T
-supportedTraceMetrics
-O
-T
-F
-F
-T
-nfDeploymentParametersList
-CM
-T
-F
-F
-T
-In addition, the ManagedFunction IOC can be associated with one or more NF Deployments needed for the particular instance of ManagedFunction.
-The NFDeploymentParameters IOC can be described as a new object in TS 28.622 [24] with the attributes below.
-Attribute Name
-S
-isReadable
-isWritable
-isInvariant
-isNotifyable
-nfDeploymentInstanceId
-M
-T
-F
-F
-T
-nfDeploymentExternalReference
-M
-T
-T
-F
-T
-Editor' s Note: It is FFS if NFDeploymentParameters IOC is to be expanded with more attributes
* * * End of Changes * * * *

S5-263622d1 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx -> S5-263622d2 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

--- S5-263622d1 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx
+++ S5-263622d2 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx
@@ -53,3 +53,4 @@
REQ-CMO-1: 3GPP management system should support LCM of Sub-Network which include 3GPP NF deployed as NF Deployments.

-REQ-CMO-A: 3GPP management system should support modelling improvements for NF Deployments.

+REQ-CMO-A: 3GPP management system should support managing multiple ManagedFunctions realized by a single NF Deployment.

+Editor’ s note: REQ-CMO-A depends on the analysis of the 6G study solutions

* * * Next Change * * * *

@@ -60,6 +61,3 @@

This key issue is related to the requirements specified in REQ-CMO-A and focusses on the following aspects

-1. How NRM can be improved to support NF Deployments?

-2. Whether and how an association is described between ManagedElement IOC and a potential NF Deployment-related IOC?

-3. Whether and how the association is described between ManagedFunction IOC and a potential NF Deployment related IOC?

-4. Whether the existing vnfParametersList attributes needs to be deprecated?

+1. Whether and how NRMs and their associations can be improved to support NF Deployments

* * * Next Change * * * *

@@ -68,5 +66,5 @@

Updates for Creation of a 3GPP Sub-network:

-The existing use case in section 5.1.19 of TS 28.531 [4] describes the end-to-end provisioning of a new 3GPP sub-network through a NF provisioning management service. Since the usage of NF Deployments is now specified in 5GA, this use case needs to limit the instantiation of NSs to ETSI NFV-MANO specific cases. Additionally, an alternative procedure would need to be defined based on other industry solutions. If necessary, additional NF Deployments for sub-networks may be created via the Deployment Management reference point.

+The existing use case in section 5.1.19 of TS 28.531 [4] describes the end-to-end provisioning of a new 3GPP sub-network through a Network provisioning management service. The Network provisioning producer exclusively uses ETSI NFV MANO network services to support this use case. Since the usage of NF Deployments is now specified in 5GA, this use case needs to limit the instantiation of NSs to ETSI NFV-MANO specific cases. Additionally, an alternative procedure would need to be defined based on other industry solutions. If necessary, additional NF Deployments for sub-networks may be created via the Deployment Management reference point.

Updates for Configuration of a 3GPP Sub-network:

-The existing use case in section 5.1.20 of TS 28.531 [4] describes the modification of 3GPP sub-network through a NF provisioning management service. Since the usage of NF Deployments is now specified in 5GA, this use case needs to limit the update of NSs to ETSI NFV-MANO specific cases. Additionally, an alternative procedure would need to be defined based on other industry solutions. If necessary, additional NF Deployments for sub-networks may be created or existing ones modified via the Deployment Management reference point.

+The existing use case in section 5.1.20 of TS 28.531 [4] describes the modification of 3GPP sub-network. The Network provisioning producer exclusively uses ETSI NFV MANO network services to support this use case. Since the usage of NF Deployments is now specified in 5GA, this use case needs to limit the update of NSs to ETSI NFV-MANO specific cases. Additionally, an alternative procedure would need to be defined based on other industry solutions. If necessary, additional NF Deployments for sub-networks may be created or existing ones modified via the Deployment Management reference point.

Editor’ s Note: It is FFS if detailed procedures are to be described for LCM of 3GPP sub-networks while using NF Deployments

S5-263622d2 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx -> S5-263622d3 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

NO TEXT CHANGE

S5-263622d3 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx -> S5-263622d5 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

--- S5-263622d3 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

+++ S5-263622d5 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

@@ -50,7 +50,6 @@

As part of the normative work in Rel-20, the usage of Deployment management reference point has been specified in TS 28.533 [5] to support lifecycle management (LCM) of NF Deployment instances. Updated use cases, requirements and procedure for LCM of NF instances realized by NF Deployments are now specified in TS 28.531 [4]. In addition, the updated procedures for Network Slice Subnet have been specified. The lifecycle impacts of 3GPP sub-networks for NF deployments, not addressed in 5G/5GA, needs to be studied as part of 6G.

-Additionally, the Network Resource Model needs to be re-examined with regards to NF Deployments. As per TS 28.662, clause 4.4.3, ManagedElement represents a telecommunication equipment which may contain hardware and software components. In the case of cloud-based instances, the ME will represent the virtualized software component only and such an instance is also known as NF Deployment. A ManagedElement may contain several managed functions. However, the ManagedElement IOC is not associated with a cloud software instance such as VNF. Currently, the vnfParametersList attribute is a part of the

ManagedFunction IOC which causes confusion and makes it difficult to associate a single VNF with multiple ManagedFunction instances. The relationship between ManagedFunction/ManagedElement and the existing VNFParametersList and the newly defined NF Deployments needs to be further examined in 6G OAM studies.

+Additionally, whether and how to update the NRM to consider NF Deployments needs to be further examined in the 6G OAM studies.

6.1.7.2.2Potential Requirements

REQ-CMO-1: 3GPP management system should support LCM of Sub-Network which include 3GPP NF deployed as NF Deployments.

-REQ-CMO-A: 3GPP management system should support managing multiple ManagedFunctions realized by a single NF Deployment.

-Editor’ s note: REQ-CMO-A depends on the analysis of the 6G study solutions

+REQ-CMO-A: 3GPP management system should support updated modelling to represent NF Deployments.

* * * Next Change * * * *

@@ -61,3 +60,3 @@

This key issue is related to the requirements specified in REQ-CMO-A and focusses on the following aspects

-1. Whether and how NRMs and their associations can be improved to support NF Deployments

+1. Whether and how Network Resource Model can be improved to support NF Deployments

* * * Next Change * * * *

S5-263622d5 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx -> 263622_final_S5-263622 pCR TR 32.801-01 LCM and NRM improvements for NF Deployments.docx

NO TEXT CHANGE

S5-263623 - 网络功能可观测性

S5-263081-converted.docx -> S5-263623 pCR 32.801-01 Cloud -- Add key issue on netework function observability agents.docx

--- S5-263081-converted.docx

+++ S5-263623 pCR 32.801-01 Cloud -- Add key issue on netework function observability agents.docx

@@ -1,44 +1,39 @@

-3GPP TSG SA5 Meeting #168S5-263081

+3GPP TSG SA5 Meeting #168S5-263623

Prague, Czech Republic, 22-26 August 2026

-Source:Hipe-IP, ETRI, LGU Plus Corp

-Title:Pseudo-CR on TR 32.801-01 CAMO - Add a key issue and potential high-level solution – Network function observability based on coordinated autonomous agents

+Source:Hipe-IP, ETRI, LGU Plus Corp, Orange

+Title:Pseudo-CR on TR 32.801-01 - Key issues on network function observability

Document for:Approval

Agenda Item:6.20.6.8

-1Decision/action requested

-The group is asked to discuss and approve the proposals.

-2References

-[1] 3GPP TR 32.801-01 v0.1.0 "Study on 6G Management and Orchestraton"

-3Rationale

-This contribution proposes to a management scenario associated with coordination of autonomous agents for cross-domain observability to TR 32.801-01 [1].

-4Detailed proposals

-1st modification

+Spec:3GPP TR 32.801-01

+Version:0.3.0

+Work Item:FS_6G_OAM

+Comments

+This pCR contains the content of the following contributions:

+S5-263081

+Pseudo-CR on TR 32.801-01 CAMO - Add a key issue and potential high-level solution – Network function observability based on coordinated autonomous agents

+Hipe-IP, ETRI, LGU Plus Corp

+Integrated in clause 7

+S5-263294

+Pseudo-CR Addition of a Key Issue and a Potential Solution for Network Functions Observability

+Orange

+Integrated in clause 7

7. 5Cloud Aspects of Management and Orchestration

7.5.1Key Issue #x: Network Functions Observability

-6G management system should support a comprehensive unified observability for network function deployments. The following dimensions are important to be considered/supported by the 3GPP management system for both Physical Network Functions (PNFs) and NFs deployed in the cloud (i.e., NF Deployments).

-1. Metrics Collection and Analysis:

+As described in scenario 6.1.7.4, the evolution towards decomposed and cloud-native network functions (NF Deployments) introduces significant observability challenges. Traditional, siloed monitoring approaches are insufficient. To meet the requirements REQ-6G-OBS-01, REQ-6G-OBS-02, and REQ-6G-OBS-03, 6G management system should support a comprehensive unified observability framework for network function deployments to ensure multi-vendor interoperability and enable end-to-end analysis.

+This key issue focuses on investigating how to define a unified observability framework within the 6G management system. The study should address the following aspects:

+1. Standardization of Observability Data: How to standardize the collection and format of key observability data (metrics, traces, logs) for both NF Deployments and Physical Network Functions (PNFs) in a consistent manner.

+2. Observability Data Collection and Analysis. How to collect and analyze the observability data

-Real-time collection of performance indicators

--Application-level metrics specific to 3GPP network functions (e.g., registration success rate, PDU session establishment time, QoS flow metrics)

--Protocol-specific metrics

-2. Distributed Tracing:

+-Real-time log streaming and historical log retrieval

+-Collection of application-level metrics specific to 3GPP network functions (e.g., registration success rate, PDU session establishment time, QoS flow metrics)

+-Collection of logs from distributed network function instances

+-Structured logging with correlation capabilities

+-Log-based anomaly detection and alerting

+3. End-to-End Correlation: Mechanisms to ensure the correlation of metrics, traces, and logs across different network functions, domains (RAN, Core), and infrastructure layers to support the following capabilities.

-End-to-end request tracing across multiple network function instances and protocols

-Capture of causality relationships between operations in distributed architectures

--Integration with industry-standard trace mechanisms (e.g., OpenTelemetry-compatible backends and OTLP protocol)

-3. Log Aggregation and Analysis:

--Collection of logs from distributed network function instances

--Structured logging with correlation capabilities

--Real-time log streaming and historical log retrieval

--Log-based anomaly detection and alerting

--Integration with metrics and traces for comprehensive diagnostics

-2nd modification

-8Solutions:

-8.X.Potential Solution #X: High-level Solution for Network Functions Observability based on Coordinated Autonomous Agents

-The Network Functions Obsevability (NFO) may support distributed observability management data processing functionalities that process the data close to its source before the data is exchanged among its functional components and with the DMFW. Observability management data from different domains may be collected by domain-specific management data collection and processing agents (e.g, RAN/CN/PNF NFO agents) and processed according to requirements provided by the consuming NFO management service.

-The NFO may also support a standardized representation of POMD exchanged between management entities. Each POMD may be associated with a POMD Descriptor that provides the information required by an MnS Consumer to correctly interpret, combine and assess the associated POMD. The POMD Descriptor may include a semantic descriptor describing the meaning and context of the POMD; a processing descriptor describing the processing operations applied, including the observation window and information loss introduced by filtering, sampling or aggregation; a quality descriptor describing characteristics such as completeness, accuracy, freshness, uncertainty and confidence; and a provenance descriptor describing the origin and processing chain of the POMD.

-A common semantic representation, potentially supported by a knowledge graph backed by NFO ontology, may be used to relate the semantic information associated with PMD from heterogeneous sources. This enables processed observability management data to be consistently interpreted and correlated by MnS Consumers. The required processing and quality characteristics may be governed by policy according to the requirements of the consuming management service.

-A NFO orchestrator (NFOO) may coordinate the processing performed by the distributed semantic processing functionalities.

-The distributed processing may support configurable processing operations, including filtering, abstraction, aggregation, normalization, correlation, enrichment or other processing operations. By performing such operations close to the data source, redundant or unnecessary management data may be removed or summarized before export, thereby reducing the volume of management data exchanged through the NFO components and the associated communication overhead. The processing may also reduce the amount of data that needs to be subsequently processed by the consuming entity, supporting management decisions within a bounded time window.

-Processing policies may determine the processing operation, processing frequency or trigger condition, data source or processing scope, and processing objective according to the requirements of the consuming management service. The processing may therefore be performed periodically or when a defined trigger condition occurs. The resulting POMD may then be provided to the MnS Consumer.

-Management data collected by management data collection and processing agents from different domains may be processed by one or more data processing functionalities before being exchanged through the DMFW. The processing may transform heterogeneous management data into Processed Management Data (PMD), which remains management data and may be consumed by MnS Consumers in the same manner as other management data. PMD may be exported periodically or upon the occurrence of a configured trigger condition according to the requirements of the consuming management service.

-Observability Analytics/Forcasting and Recommendation agents may support capabilities of detecting/predicting anomalous events and recommending any resolution policies based on the POMD collected and processed by other agents. The coordination among these agents are supported by the NFOO.

-Fig. 6.1.4.x.1.1 Network function observability based on coordinated autonomous agents

-End of modifications

+-Integration of metrics, traces, and logs for comprehensive diagnostics

+4. Observability Control and Management: Definition of control and management capabilities to allow an MnS consumer to configure and control the observability framework including the following capabilities.

+-Setting data collection policies, sampling rates for traces

+-Defining specific metrics to be collected.

+5. Integration with Industry Standards: How to leverage and integrate de-facto industry standards, such as OpenTelemetry, within the 3GPP Service-Based Management Architecture (SBMA) to facilitate adoption and avoid reinventing existing mechanisms.

S5-263623 pCR 32.801-01 Cloud -- Add key issue on netework function observability agents.docx -> 263623_final_S5-263623 pCR 32.801-01 Cloud -- Add key issue on netework function observability agents.docx

NO TEXT CHANGE

S5-263624 - Cloud管理框架

S5-263110 Pseudo-CR on TR 32.801-01 add Key Issues for Cloud Aspects Management Framework.docx -> S5-263624d1 Pseudo-CR on TR 32.801-01 add Key Issues for Cloud Aspects Management Framework.docx

--- S5-263110 Pseudo-CR on TR 32.801-01 add Key Issues for Cloud Aspects Management Framework.docx

+++ S5-263624d1 Pseudo-CR on TR 32.801-01 add Key Issues for Cloud Aspects Management Framework.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263110

+3GPP TSG SA5 Meeting #168S5-263624

Prague, Czech Republic, 24-28 August 2026

@@ -13,6 +13,6 @@

* * * First Change * * * *

-XCloud Aspects of Management and Orchestration

-7.X.1 Key Issue #X: Architectural Framework for Cloud Aspects Management

+5Cloud Aspects of Management and Orchestration

+7.5.X Key Issue #X: Architectural Framework for Cloud Aspects Management

To support the 6G management system in adopting cloud-native principles for cloud aspects of management, enhancements to the existing management architecture are needed, including the evolution of management capabilities and coordination mechanisms with external cloud orchestration and management systems. This key issues focuses on the following aspects:

--Evolution and enhancement of management architecture: whether and how the existing 3GPP management framework needs to be enhanced, and identify whether MnFs and MnSs that need to be newly introduced or enhanced in 6G.

+-Evolution and enhancement of management architecture: whether and how the existing 3GPP management framework as defined in clause 5.2 of TS 28.533[5] needs to be enhanced, and identify whether MnFs and MnSs that need to be newly introduced or enhanced in 6G.

-Interaction mechanisms with external cloud orchestration and management systems: identify the interaction between the 6G management system and external cloud orchestration and management systems, as well as the enhancement and evolution of interfaces such as the deployment management reference point.

S5-263624d1 Pseudo-CR on TR 32.801-01 add Key Issues for Cloud Aspects Management Framework.docx -> 263624_final_S5-263624 Pseudo-CR on TR 32.801-01 add Key Issues for Cloud Aspects Management Framework.docx

NO TEXT CHANGE

S5-263625 - 声明式LCM需求与关键问题

Pseudo-CR-263295-Add gitops use case solutions and key issue for OAM.docx -> S5-263625d1-(was263295)-Add gitops use case solutions and key issue for OAM.docx

--- Pseudo-CR-263295-Add gitops use case solutions and key issue for OAM.docx

+++ S5-263625d1-(was263295)-Add gitops use case solutions and key issue for OAM.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263295

+3GPP TSG SA5 Meeting #168S5-263625d1

Prague, Czech Republic, 24-28 August 2026

@@ -11,3 +11,3 @@

Abstract:

-This contribution proposes a new Key Issue and a corresponding Potential Solution to address the integration of GitOps-based lifecycle management (LCM) with the 3GPP management system.

+This contribution proposes a new Key Issue and a corresponding Potential Solution to address the integration of Declarative-based lifecycle management (LCM) with the 3GPP management system.

Building upon the management scenario described in clause 6.1.7.3 of TR 32.801, this contribution first introduces a Key Issue that highlights the challenge of integrating the 3GPP Service-Based Management Architecture (SBMA) with external GitOps orchestration systems.

@@ -17,82 +17,12 @@

6.1.7.3.2Potential Requirements

-REQ-6G-GITOPS-03: The 6G management system should support managing NF deployment through declarative approach.

+REQ-6G-GITOPS-03: The 6G management system should support managing NF through declarative approach.

* * * Next Change * * * *

-7.5.x Key Issue #X: Integration of GitOps-based LCM with the 3GPP Service-Based Management Architecture

-As described in scenario 6.1.7.3, a GitOps-based approach offers a robust, declarative, and version-controlled framework for the lifecycle management (LCM) of NF Deployments. However, GitOps mechanisms (e.g., Git repositories, CI/CD pipelines, GitOps controllers like Flux or ArgoCD) typically operate externally to the 3GPP management system.

-A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-GITOPS-01 and REQ-6G-GITOPS-02. The editor's note in clause 6.1.7.3 highlights the need to describe this relationship with the SBMA.

-This key issue focuses on investigating the architectural integration between the 3GPP SBMA and external GitOps-based management and orchestration systems. The study should address the following aspects:

-1. Standardized Interface for Declarative LCM: How to define a standardized management service within the SBMA that allows an MnS consumer to manage the desired state of an NF Deployment declaratively, without being exposed to the underlying GitOps implementation details.

-2. Mapping 3GPP Operations to GitOps Workflows: How to map 3GPP LCM operations (e.g., instantiation, modification, termination of an NF Deployment) to corresponding GitOps actions (e.g., committing a declarative manifest to a Git repository and triggering a reconciliation loop) and vice-versa.

-3. State Reconciliation and Feedback Mechanisms: How the 3GPP management system can receive feedback and notifications about the status of a GitOps-driven operation. This includes reporting on the progress of the deployment pipeline, the reconciliation status (e.g., `InProgress`, `Succeeded`, `Failed`), and providing auditable traceability information as reqested in REQ-6G-GITOPS-02.

-4. Information Model for Desired State: How to model the "desired state" of an NF Deployment. This involves defining how 3GPP NRM attributes and parameters can be represented in a format suitable for GitOps workflows (e.g., YAML manifests, Helm charts), while maintaining consistency with 3GPP information models.

+7.5.x Key Issue #X: Integration of declarative-based LCM with the 3GPP Service-Based Management Architecture

+As described in scenario 6.1.7.3, a declarative-based approach offers a robust, declarative, and version-controlled framework for the lifecycle management (LCM) of NF. However, GitOps mechanisms (e.g., Git repositories, CI/CD pipelines, GitOps controllers like Flux or ArgoCD) typically operate externally to the 3GPP management system.

+A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-GITOPS-01, REQ-6G-GITOPS-02 and REQ-6G-GITOPS-03. The editor's note in clause 6.1.7.3 highlights the need to describe this relationship with the SBMA.

+This key issue focuses on investigating the architectural integration between the 3GPP SBMA and external declarative-based management and orchestration systems. The study should address the following aspects:

+1. Standardized Interface for Declarative LCM: How to define a standardized management service within the SBMA that allows an MnS consumer to manage the desired state of a declaratively managed NF Deployment, without being exposed to the underlying GitOps implementation details.

+2. Mapping 3GPP Operations to GitOps Workflows and vice-versa: How to map 3GPP LCM operations (e.g., instantiation, modification, termination of an NF Deployment) to corresponding declarative-based actions (e.g., committing a declarative manifest to a Git repository and triggering a reconciliation loop).

+3. State Reconciliation and Feedback Mechanisms: How the 3GPP management system can receive feedback and notifications about the status of a declarative-based driven operation. This includes reporting on the progress of the deployment pipeline, the reconciliation status (e.g., `InProgress`, `Succeeded`, `Failed`), and providing auditable traceability information as reqested in REQ-6G-GITOPS-02.

+4. Information Model for Desired State: How to model the "desired state" of a declaratively managed NF Deployment. This involves defining how 3GPP NRM attributes and parameters can be represented in a format suitable for declarative-based workflows (e.g., YAML manifests, Helm charts), while maintaining consistency with 3GPP information models.

* * * Next Change * * * *

-8.x Potential Solution #X: A Management Service for GitOps-based NF Deployment LCM

-8.x.1 Description

-This potential solution addresses the key issue 1 on integrating GitOps-based LCM with the SBMA (7.5.x). It proposes the introduction of a new Management Service, the Declarative Lcm MnS, which acts as a standardized abstraction layer between the 3GPP management system and external GitOps-based orchestration systems.

-This approach aligns with the principles of cloud management), where the underlying infrastructure capabilities are exposed via declarative, intent-like APIs. The `DeclarativeLcmMnS` shields the MnS consumer from the complexity of the GitOps toolchain (Git, CI/CD, controllers) and provides a single, standardized point of interaction.

-8.x.2 Key Principles and Components

-The solution is based on the following principles:

-1. Abstraction via a Dedicated MnS: An authorized MnS consumer (e.g., a Network Slice Management Function) interacts with the DeclarativeLcmMnS producer exclusively through its standardized management service interface. This interface provides a complete abstraction of the underlying GitOps toolchain, shielding the consumer from all implementation-specific details such as the Git repository or the GitOps controller.

-2. Declarative Operations: The MnS consumer provides the desired state of the NF Deployment as a declarative payload in its management request. This payload contains the necessary parameters, which the MnS producer then translates into the appropriate format for the GitOps system (e.g., a Helm chart `values.yaml` file).

-3. MnS Producer as the GitOps mediator: The `DeclarativeLcmMnS` producer is responsible for the entire GitOps workflow on behalf of the 3GPP management system. Its responsibilities include:

-* Authenticating with the Git repository.

-* Generating and validating the declarative manifests (e.g., YAML files).

-* Committing the new desired state to the Git repository, which serves as the single source of truth.

-* Monitoring the external GitOps controller for the reconciliation status.

-4. Standardized Feedback Loop: The `DeclarativeLcmMnS` producer observes the state of the deployment in the cloud-native environment (as reported by the GitOps controller) and reports it back to the MnS consumer using standard 3GPP mechanisms, such as notifications (`notifyStateChange`) or queryable attributes. This provides the traceability and auditability required by **REQ-6G-GITOPS-02**.

-8.x.3 High-Level Architecture

-The figure below illustrates the proposed architecture.

-Figure 8.x.3-1: High-level architecture for GitOps-based LCM Integration

-Flow Description:

-1. The MnS Consumer invokes a declarative LCM operation (e.g., `createNfDeployment`) to the DeclarativeLcmMnS Producer, including the desired state specification.

-2. The DeclarativeLcmMnS Producer translates the request into a declarative artifact (e.g., a set of YAML files or a Helm chart update).

-3. The MnS Producer commits and pushes the artifact to the Git Repository.

-4. The GitOps Controller (e.g., ArgoCD, Flux), which is monitoring the repository, detects the change in the desired state.

-5. The GitOps Controller pulls the artifact and applies the desired state to the Cloud Platform (e.g., Kubernetes), triggering the instantiation or modification of the NF Deployment.

-6. The GitOps Controller continuously reconciles the actual state with the desired state and updates its own status.

-7. The DeclarativeLcmMnS Producer monitors the status reported by the GitOps Controller.

-8. The MnS Producer sends a notification to the MnS Consumer to report the final status of the operation (e.g., `Completed`, `Failed`) and provides a link to the Git commit for full traceability.

-8.x.4 Mapping to Key Issues

-This solution directly addresses Key Issue 7.5.x by providing a concrete architectural proposal that integrates a GitOps-based workflow into the 3GPP SBMA. It defines a clear, standardized interface (`DeclarativeLcmMnS`) that abstracts the underlying implementation, ensures a reliable feedback loop for state reconciliation, and maintains the Git repository as the single source of truth, thus fulfilling the requirements of the management scenario.

```
-* * * Next Change * * * *
-Annex X: UML of High-level architecture for GitOps-based LCM Integration
-@startuml
-!theme plain
-title GitOps-based LCM Integration with 3GPP Management
-' Define packages for logical grouping
-package "3GPP Management System" {
-actor "MnS Consumer" as Consumer
-component "DeclarativeLcmMnS\nProducer" as Producer
-}
-package "External GitOps Orchestration System" {
-database "Git Repository\n(Source of Truth)" as GitRepo
-component "GitOps Controller\n(e.g., ArgoCD, Flux)" as Controller
-node "Cloud Platform\n(e.g., Kubernetes)" as Platform
-}
-' --- Flow Description ---
-' Step 1: Consumer sends request
-Consumer -> Producer : 1. Declarative LCM Request
-' Step 2 & 3: Producer translates and commits
-Producer -> GitRepo : 2. & 3. Translate request & Commit artifact
-note on link
-The Producer translates the declarative
-request into a GitOps-compatible
-artifact (e.g., YAML) and commits it.
-end note
-' Step 4: Controller detects the change
-Controller ..> GitRepo : 4. Detects change in desired state
-' Step 5: Controller applies the change
-Controller -> Platform : 5. Pulls artifact & applies desired state
-' Step 6: Controller reconciles the state and updates its status
-Platform ..> Controller : 6. Reconciles actual state & updates status
-note on link
-The controller continuously monitors the
-platform to ensure the actual state
-matches the desired state and updates its own status.
-end note
-' Step 7: Producer monitors the controller's status
-Producer ..> Controller : 7. Monitors status reported by Controller
-' Step 8: Producer sends notification back to the consumer
-Producer --> Consumer : 8. Sends notification (e.g., Completed/Failed)
-@enduml
-* * * End of Changes * * * *
```

S5-263625d1-(was263295)-Add gitops use case solutions and key issue for OAM.docx -> S5-263625d2-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

```
--- S5-263625d1-(was263295)-Add gitops use case solutions and key issue for OAM.docx
+++ S5-263625d2-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx
@@ -1,2 +1,2 @@
-3GPP TSG SA5 Meeting #168S5-263625d1
+3GPP TSG SA5 Meeting #168S5-263625d2
```

Prague, Czech Republic, 24-28 August 2026

S5-263625d2-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx -> S5-263625d3-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

--- S5-263625d2-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

+++ S5-263625d3-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263625d2

+3GPP TSG SA5 Meeting #168S5-263625d3

Prague, Czech Republic, 24-28 August 2026

@@ -17,3 +17,3 @@

6.1.7.3.2Potential Requirements

-REQ-6G-GITOPS-03: The 6G management system should support managing NF through declarative approach.

+REQ-6G-GITOPS-03: The 6G management system should support managing NF deployment through declarative approach.

* * * Next Change * * * *

S5-263625d3-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx -> S5-263625d4-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

--- S5-263625d3-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

+++ S5-263625d4-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263625d3

+3GPP TSG SA5 Meeting #168S5-263625d4

Prague, Czech Republic, 24-28 August 2026

@@ -20,3 +20,3 @@

7.5.x Key Issue #X: Integration of declarative-based LCM with the 3GPP Service-Based Management Architecture

-As described in scenario 6.1.7.3, a declarative-based approach offers a robust, declarative, and version-controlled framework for the lifecycle management (LCM) of NF. However, GitOps mechanisms (e.g., Git repositories, CI/CD pipelines, GitOps controllers like Flux or ArgoCD) typically operate externally to the 3GPP management system.

+As described in scenario 6.1.7.3, a declarative-based approach offers a robust, declarative, and version-controlled framework for the lifecycle management (LCM) of NF Deployments. However, GitOps mechanisms (e.g., Git repositories, CI/CD pipelines, GitOps controllers like Flux or ArgoCD) typically operate externally to the 3GPP management system.

A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-GITOPS-01, REQ-6G-GITOPS-02 and REQ-6G-GITOPS-03. The editor's note in clause 6.1.7.3 highlights the need to describe this relationship with the SBMA.

S5-263625d4-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx -> S5-263625d5-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

--- S5-263625d4-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

+++ S5-263625d5-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263625d4

+3GPP TSG SA5 Meeting #168S5-263625d5

Prague, Czech Republic, 24-28 August 2026

@@ -21,8 +21,8 @@

As described in scenario 6.1.7.3, a declarative-based approach offers a robust, declarative, and version-controlled framework for the lifecycle management (LCM) of NF Deployments. However, GitOps mechanisms (e.g., Git repositories, CI/CD pipelines, GitOps controllers like Flux or ArgoCD) typically operate externally to the 3GPP management system.

-A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-GITOPS-01, REQ-6G-GITOPS-02 and REQ-6G-GITOPS-03. The editor's note in clause 6.1.7.3 highlights the need to describe this relationship with the SBMA.

+A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-GITOPS-01, REQ-6G-GITOPS-02 and REQ-6G-GITOPS-03.

This key issue focuses on investigating the architectural integration between the 3GPP SBMA and external declarative-based management and orchestration systems. The study should address the following aspects:

-1. Standardized Interface for Declarative LCM: How to define a standardized management service within the SBMA that allows an MnS consumer to manage the desired state of a declaratively managed NF Deployment, without being exposed to the underlying GitOps implementation details.

-2. Mapping 3GPP Operations to GitOps Workflows and vice-versa: How to map 3GPP LCM operations (e.g., instantiation, modification, termination of an NF Deployment) to corresponding declarative-based actions (e.g., committing a declarative manifest to a Git repository and triggering a reconciliation loop).

-3. State Reconciliation and Feedback Mechanisms: How the 3GPP management system can receive feedback and notifications about the status of a declarative-based driven operation. This includes reporting on the progress of the deployment pipeline, the reconciliation status (e.g., `InProgress`, `Succeeded`, `Failed`), and providing auditable traceability information as reqested in REQ-6G-GITOPS-02.

-4. Information Model for Desired State: How to model the "desired state" of a declaratively managed NF Deployment. This involves defining how 3GPP NRM attributes and parameters can be represented in a format suitable for declarative-based workflows (e.g., YAML manifests, Helm charts), while maintaining consistency with 3GPP information models.

+1. Standardized Interface for Declarative LCM: How to enable a standardized management service within the SBMA to determine the desired state of a declaratively managed NF Deployment, without being exposed to the underlying implementation details.

+2. Mapping 3GPP Operations to declarative-based Workflows and vice-versa: How to map 3GPP MnS provisioning service operations (e.g., creation, modification, termination of an NF) to corresponding declarative-based actions for an NF deployment (e.g., committing a declarative manifest to a Git repository and triggering a reconciliation loop).

+3. State Reconciliation and Feedback Mechanisms: How the 3GPP management system can receive feedback and notifications about the status of a declarative-based driven operation. This includes reporting on the progress of the deployment pipeline, the reconciliation status (e.g., `InProgress`, `Succeeded`, `Failed`), and providing auditable traceability information as requested in REQ-6G-GITOPS-02.

+4. Information Model for Desired State: How to determine the "desired state" of a declaratively managed NF Deployment. This involves defining how 3GPP NRM attributes and parameters can be represented in a format suitable for declarative-based workflows (e.g., YAML manifests, Helm charts), while maintaining consistency with 3GPP information models.

* * * Next Change * * * *

S5-263625d5-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx -> S5-263625d6-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

--- S5-263625d5-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

+++ S5-263625d6-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263625d5

+3GPP TSG SA5 Meeting #168S5-263625d6

Prague, Czech Republic, 24-28 August 2026

@@ -13,3 +13,2 @@

Building upon the management scenario described in clause 6.1.7.3 of TR 32.801, this contribution first introduces a Key Issue that highlights the challenge of integrating the 3GPP Service-Based Management Architecture (SBMA) with external GitOps orchestration systems.

-Subsequently, a Potential Solution is proposed, centered around the creation of a new Management Service, the DeclarativeLcmMnS. This MnS is designed to act as a standardized abstraction layer, shielding MnS consumers from the underlying GitOps toolchain and providing a formal, traceable mechanism for the declarative management of NF Deployments.

Proposed Changes

@@ -17,3 +16,3 @@

6.1.7.3.2Potential Requirements

-REQ-6G-GITOPS-03: The 6G management system should support managing NF deployment through declarative approach.

+REQ-6G-DECL-03: The 6G management system should support managing NF deployment through declarative approach.

* * * Next Change * * * *

@@ -21,3 +20,3 @@

As described in scenario 6.1.7.3, a declarative-based approach offers a robust, declarative, and version-controlled framework for the lifecycle management (LCM) of NF Deployments. However, GitOps mechanisms (e.g., Git repositories, CI/CD pipelines, GitOps controllers like Flux or ArgoCD) typically operate externally to the 3GPP management system.

-A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-GITOPS-01, REQ-6G-GITOPS-02 and

REQ-6G-GITOPS-03.

+A key challenge is to define how the 3GPP management system can interwork with these external GitOps systems in a standardized, secure, and reliable manner, fulfilling requirements REQ-6G-DECL-01, REQ-6G-DECL-02 and REQ-6G-DECL-03.

This key issue focuses on investigating the architectural integration between the 3GPP SBMA and external declarative-based management and orchestration systems. The study should address the following aspects:

@@ -25,4 +24,4 @@

2. Mapping 3GPP Operations to declarative-based Workflows and vice-versa: How to map 3GPP MnS provisioning service operations (e.g., creation, modification, termination of an NF) to corresponding declarative-based actions for an NF deployment (e.g., committing a declarative manifest to a Git repository and triggering a reconciliation loop).

-3. State Reconciliation and Feedback Mechanisms: How the 3GPP management system can receive feedback and notifications about the status of a declarative-based driven operation. This includes reporting on the progress of the deployment pipeline, the reconciliation status (e.g., `InProgress`, `Succeeded`, `Failed`), and providing auditable traceability information as requested in REQ-6G-GITOPS-02.

-4. Information Model for Desired State: How to determine the "desired state" of a declaratively managed NF Deployment. This involves defining how 3GPP NRM attributes and parameters can be represented in a format suitable for declarative-based workflows (e.g., YAML manifests, Helm charts), while maintaining consistency with 3GPP information models.

+3. State Reconciliation and Feedback Mechanisms: How the 3GPP management system can receive feedback and notifications about the status of a declarative-based driven operation. This includes reporting on the progress of the deployment pipeline, the reconciliation status (e.g., `InProgress`, `Succeeded`, `Failed`), and providing auditable traceability information as requested in REQ-6G-DECL-02.

+4. Information Model for Desired State: How to determine the "desired state" of a declaratively managed NF Deployment.

* * * Next Change * * * *

S5-263625d6-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx -> 263625_final_S5-263625-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

--- S5-263625d6-(was263295)-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

+++ 263625_final_S5-263625-pCR TR 32.801.01 Add gitops use case solutions and key issue for OAM.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263625d6

+3GPP TSG SA5 Meeting #168S5-263625

Prague, Czech Republic, 24-28 August 2026

@@ -26,2 +26,2 @@

4. Information Model for Desired State: How to determine the "desired state" of a declaratively managed NF Deployment.

-* * * Next Change * * * *

+* * * End of Changes * * * *

S5-263626 - 主动韧性准备与管理功能部署

S5-263296-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions.docx -> S5-263626d1-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx

--- S5-263296-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions.docx

+++ S5-263626d1-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263296

+3GPP TSG SA5 Meeting #168S5-263626d1

Prague, Czech Republic, 24-28 August 2026

@@ -10,3 +10,3 @@

Comments

-Abstract: This contribution proposes additional potential requirements for the existing management scenario on General Cloud Aspects Use Case for 6G Resiliency (clause 6.1.7.1). The existing requirements address reacting to NF Deployment issues; the proposed additions address preparing for them: the derivation of fallback placement and recovery configurations for NF Deployments before failures occur, and the resilient placement of the management functions themselves, so that management and recovery capabilities remain available when parts of the underlying infrastructure fail.

+Abstract: This contribution proposes two additional potential requirements for the management scenario on General Cloud Aspects Use Case for 6G Resiliency (clause 6.1.7.1): the proactive derivation of fallback homing and recovery configurations for NF Deployments and their use when requesting their reallocation upon failure detection (REQ-Cloud-X), and the redundant or relocatable deployment of the Management Functions of the 3GPP management system that are themselves deployed on cloudified/virtualized infrastructures (REQ-Cloud-Y). This is a revision of S5-263296 addressing the comments received; the changes are shown with revision marks relative to the initial version. The requirement numbers are left as REQ-Cloud-X and REQ-Cloud-Y for final numbering by the rapporteur.

Proposed Changes

@@ -16,5 +16,7 @@

REQ-Cloud-2: The 6G management system should be able to configure and request external orchestration and management systems to automatically reallocate NF Deployment to maintain service continuity and performance.

-REQ-Cloud-3: The 6G management system should support the derivation of fallback placement and recovery configurations for NF Deployments before failures occur (e.g., based on the evaluation of failure scenarios), so that reallocation decisions in failure situations can be taken and executed quickly.

-REQ-Cloud-4: The 6G management system should support the resilient placement of management functions (e.g., redundant or relocatable deployment across sites), so that management and recovery capabilities remain available when parts of the underlying infrastructure fail.

-NOTE: The relation to the disaster resilience aspects of the general management scenarios in clause 6.1.1 is FFS.

+REQ-Cloud-X:: The 6G management system should have the capability to derive, before failures occur, fallback homing and recovery configurations for NF Deployments (e.g., candidate target clusters or sites, derived from the evaluation of failure scenarios such as the failure or degradation of a cluster or site hosting NF Deployments), and the capability to use the pre-derived configurations when requesting external orchestration and management systems to reallocate the affected NF Deployments to a different target cluster or site upon detection of such a failure (e.g., based on the information obtained according to REQ-Cloud-1).

+REQ-Cloud-Y: The 6G management system should have the capability to deploy Management Functions of the 3GPP management system that are themselves deployed on cloudified/virtualized infrastructures in a redundant or relocatable manner across different sites or clusters, including the capability to interact with external orchestration and management systems for this purpose.

+NOTE 1: REQ-Cloud-X and REQ-Cloud-Y refer to the selection of the target cluster or site for an NF Deployment as a whole (referred to as homing in some organizations, e.g., O-RAN). The scheduling of workloads onto worker nodes within a cluster is internal to the cluster and is not addressed by these requirements.

+NOTE 2: The capabilities in REQ-Cloud-X and REQ-Cloud-Y are provided on the MnS producer side of the 6G management system; no behaviour is required from MnS consumers. The pre-derived fallback configurations in REQ-Cloud-X are retained by the 3GPP management system, and the execution of the reallocation remains with the applicable external orchestration and management systems, consistent with REQ-Cloud-2. REQ-Cloud-X addresses the NF Deployments managed by the 6G management system, whereas REQ-Cloud-Y addresses the Management Functions of the 3GPP management system itself (e.g., MnS producers), whose deployment on cloudified/virtualized infrastructures follows the cloud-native principle in clause 5.2.1. Which Management Functions require redundant or relocatable deployment is a deployment decision and is not specified.

+Editor’ s note: Alignment with related ongoing work in other SDOs (e.g., the homing concept in O-RAN) to avoid conflicting specifications is FFS.

* * * End of Changes * * * *

S5-263626d1-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx -> S5-263626d2-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx

--- S5-263626d1-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx

+++ S5-263626d2-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263626d1

+3GPP TSG SA5 Meeting #168S5-263626d2

Prague, Czech Republic, 24-28 August 2026

@@ -16,7 +16,4 @@

REQ-Cloud-2: The 6G management system should be able to configure and request external orchestration and management systems to automatically reallocate NF Deployment to maintain service continuity and performance.

-REQ-Cloud-X:: The 6G management system should have the capability to derive, before failures occur, fallback homing and recovery configurations for NF Deployments (e.g., candidate target clusters or sites, derived from the evaluation of failure scenarios such as the failure or degradation of a cluster or site hosting NF Deployments), and the capability to use the pre-derived configurations when requesting external orchestration and management systems to reallocate the affected NF Deployments to a different target cluster or site upon detection of such a failure (e.g., based on the information obtained according to REQ-Cloud-1).

-REQ-Cloud-Y: The 6G management system should have the capability to deploy Management Functions of the 3GPP management system that are themselves deployed on cloudified/virtualized infrastructures in a redundant or relocatable manner across different sites or clusters, including the capability to interact with external orchestration and management systems for this purpose.

-NOTE 1: REQ-Cloud-X and REQ-Cloud-Y refer to the selection of the target cluster or site for an NF Deployment as a whole (referred to as homing in some organizations, e.g., O-RAN). The scheduling of workloads onto worker nodes within a cluster is internal to the cluster and is not addressed by these requirements.

-NOTE 2: The capabilities in REQ-Cloud-X and REQ-Cloud-Y are provided on the MnS producer side of the 6G management system; no behaviour is required from MnS consumers. The pre-derived fallback configurations in REQ-Cloud-X are retained by the 3GPP management system, and the execution of the reallocation remains with the applicable external orchestration and management systems, consistent with REQ-Cloud-2. REQ-Cloud-X addresses the NF Deployments managed by the 6G management system, whereas REQ-Cloud-Y addresses the Management Functions of the 3GPP management system itself (e.g., MnS producers), whose deployment on cloudified/virtualized infrastructures follows the cloud-native principle in clause 5.2.1. Which Management Functions require redundant or relocatable deployment is a deployment decision and is not specified.

-Editor’ s note: Alignment with related ongoing work in other SDOs (e.g., the homing concept in O-RAN) to avoid conflicting specifications is FFS.

+REQ-Cloud-X: The 6G management system should have the capability, based on acquired NF Deployment information, telemetry information, and virtualized/cloudified infrastructure location and capacity, to compute and derive location requirements for NF Deployment (re-)allocation, including deriving such location requirements in advance for potential failure situations, to support NF service continuity.

+REQ-Cloud-Y: The 6G management system should have the capability to request external orchestration and management systems to deploy the Management Functions of the 3GPP management system that are hosted on cloudified/virtualized infrastructures in a redundant or relocatable manner across different sites.

* * * End of Changes * * * *

S5-263626d2-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx -> 263626_final_S5-263626-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions.docx

--- S5-263626d2-(was263296)-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions_v4.docx

+++ 263626_final_S5-263626-PseudoCR TR 32.801-01 Add requirements on proactive resilience and resilient placement of management functions.docx

@@ -1,2 +1,2 @@

-3GPP TSG SA5 Meeting #168S5-263626d2

+3GPP TSG SA5 Meeting #168S5-263626

Prague, Czech Republic, 24-28 August 2026